US012402312B2

(12) **United States Patent**
Choi

(10) **Patent No.: US 12,402,312 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE**

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(72) Inventor: **Daehwan Choi**, Suwon-si (KR)

(73) Assignee: **Samsung Electronics Co., Ltd.**, Gyeonggi-Do (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 525 days.

(21) Appl. No.: **17/954,968**

(22) Filed: **Sep. 28, 2022**

(65) **Prior Publication Data**

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(30) **Foreign Application Priority Data**

Jan. 4, 2022 (KR) 10-2022-0000680

(51) **Int. Cl.**
H10B 43/27 (2023.01)
H10B 41/10 (2023.01)
(Continued)

(52) **U.S. Cl.**
CPC ***H10B 43/27*** (2023.02); ***H10B 41/10*** (2023.02); ***H10B 41/27*** (2023.02); ***H10B 41/35*** (2023.02); ***H10B 43/10*** (2023.02); ***H10B 43/35*** (2023.02)

(58) **Field of Classification Search**
CPC H10B 43/27; H10B 41/10; H10B 41/27; H10B 41/35; H10B 43/10; H10B 43/35;
(Continued)

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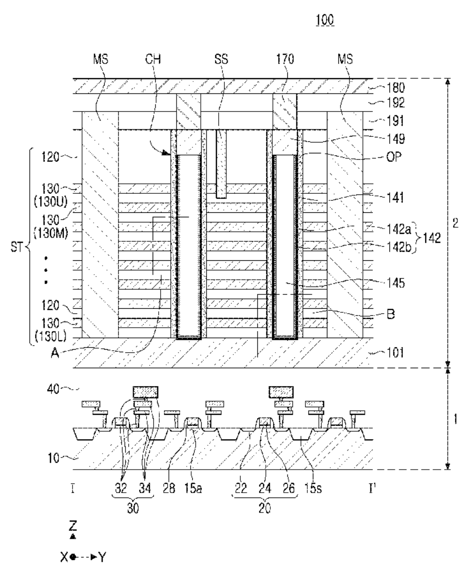
Primary Examiner — David Vu
(74) *Attorney, Agent, or Firm* — Harness, Dickey & Pierce, P.L.C.

(57) **ABSTRACT**

A method of manufacturing a semiconductor device includes forming a stacked structure by stacking gate layers and interlayer insulating layers alternately on a substrate; and forming a channel structure passing through the stacked structure in a vertical direction, wherein the forming a channel structure includes forming an opening by etching the stacked structure; forming a gate insulating layer covering a side surface of the opening; forming a variable resistive material layer on the gate insulating layer; changing an oxygen vacancy concentration in a region of the variable resistive material layer by performing a plasma treatment process or an annealing process on the variable resistive material layer; forming a core insulating pattern covering the variable resistive material layer and filling at least a portion of the opening; and forming a pad pattern on the core insulating pattern.

20 Claims, 29 Drawing Sheets



(51) **Int. Cl.**

H10B 41/27	(2023.01)
H10B 41/35	(2023.01)
H10B 43/10	(2023.01)
H10B 43/35	(2023.01)
H10N 70/00	(2023.01)

(58) **Field of Classification Search**

CPC H10B 63/30; H10B 63/34; H10B 63/845; H10B 43/50; H10D 64/037; H10N 70/041; H10N 70/24; H10N 70/826; H10N 70/8833; H10N 70/8836

See application file for complete search history.

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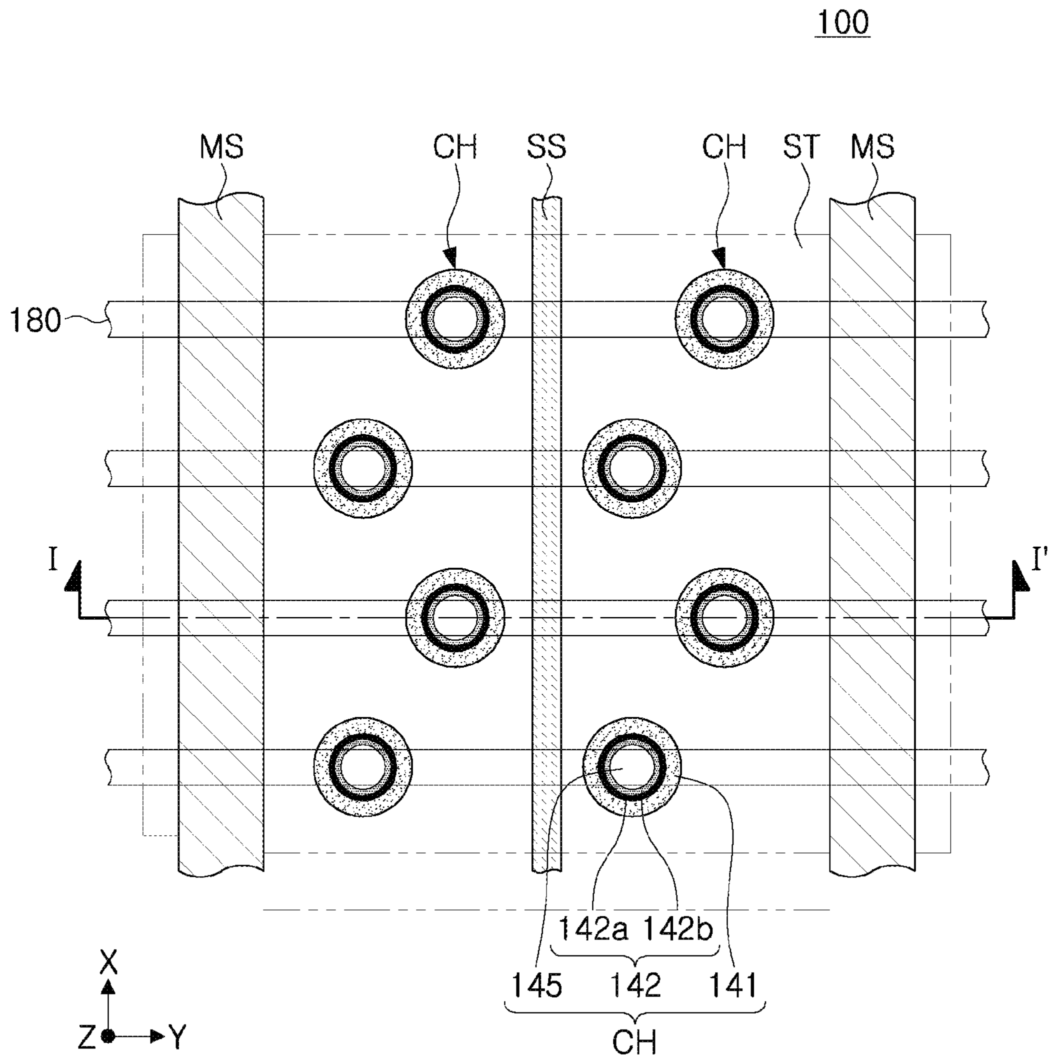


FIG. 1

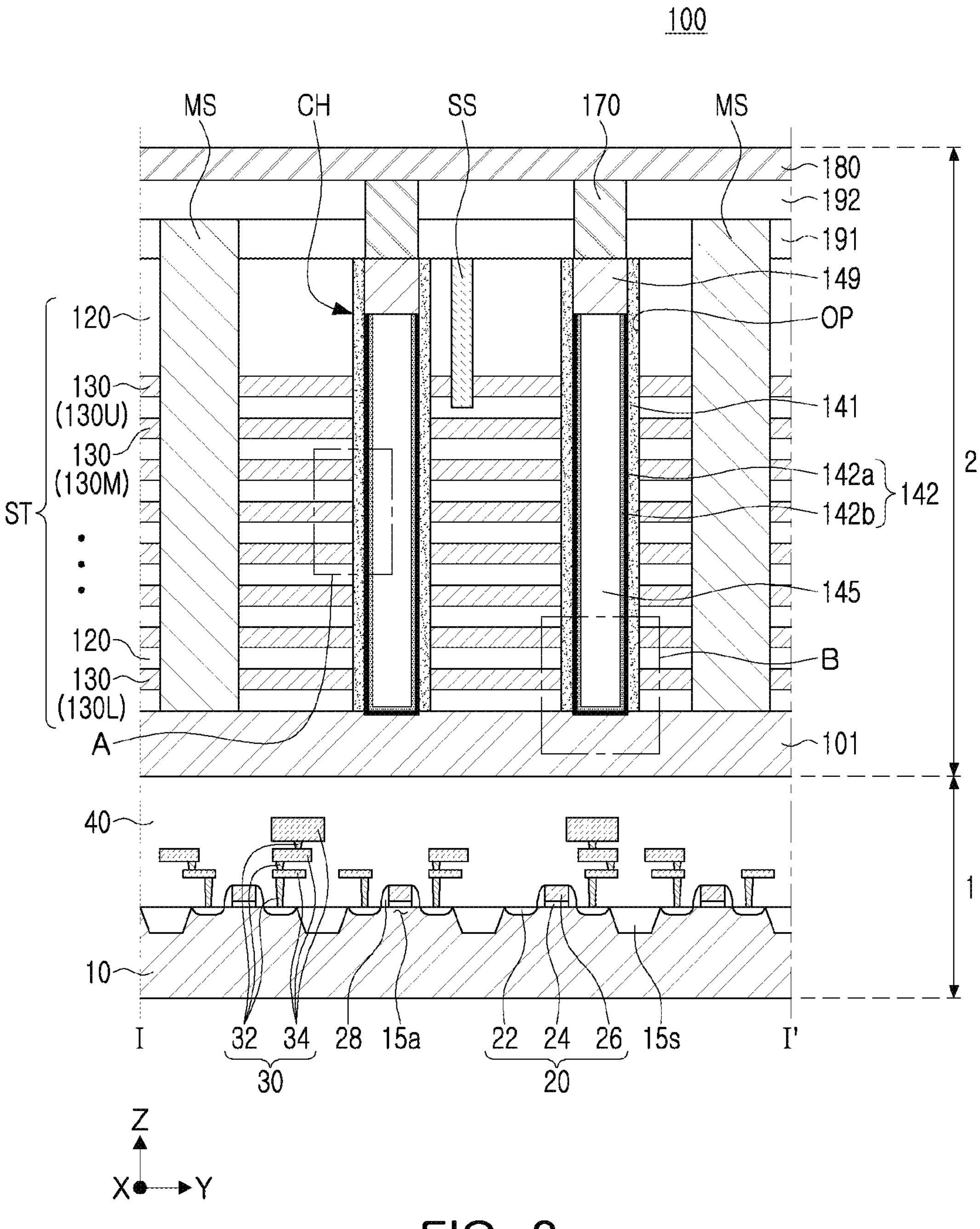

100
MS
CH
SS
170
MS
180
192
191
149
OP
141
142a
142b
142
145
B
101
2
1
120
130
(130U)
130
(130M)
ST
120
130
(130L)
A
40
10
I
32
34
30
28
15a
22
24
26
20
15s
I'
Z
X
Y

FIG. 2

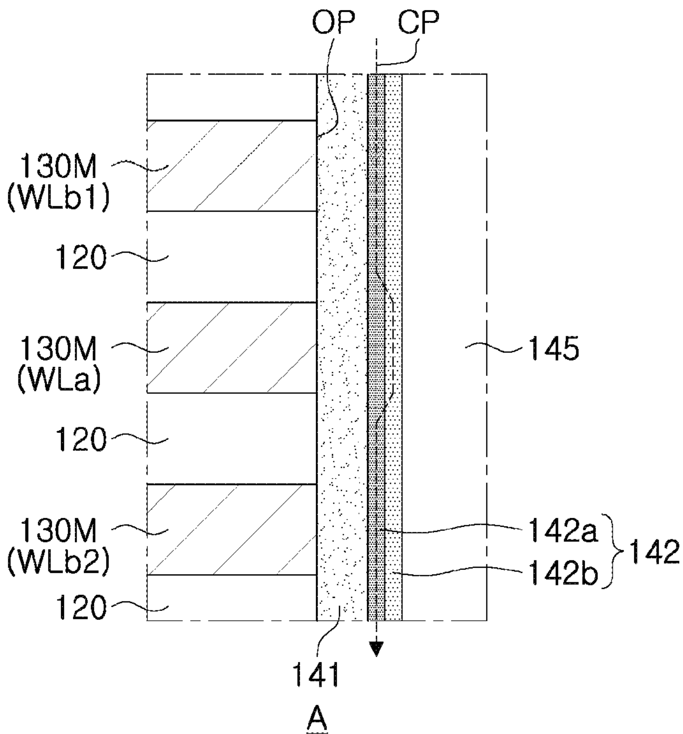

OP
CP
130M
(WLb1)
120
130M
(WLa)
145
120
130M
(WLb2)
142a
142
142b
120
141
A

FIG. 3

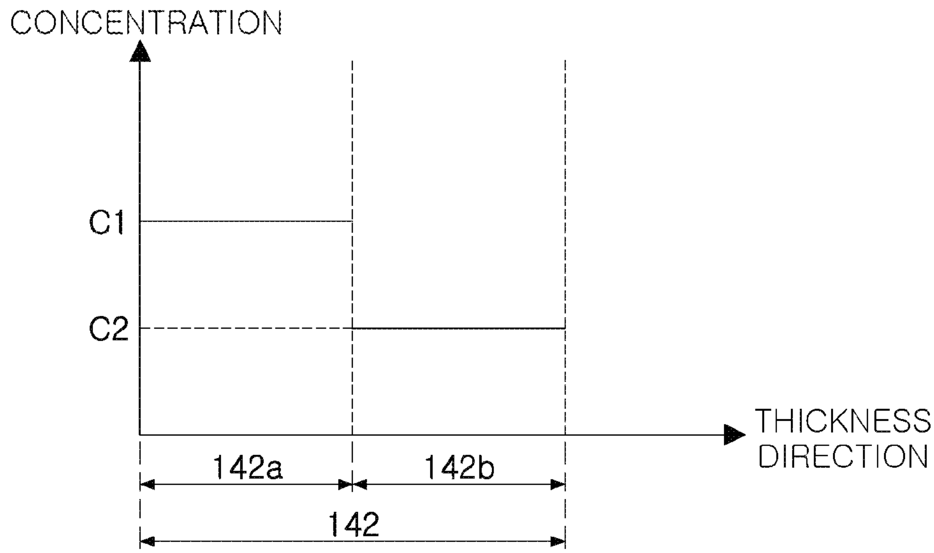

CONCENTRATION
C1
C2
THICKNESS DIRECTION
142a
142b
142

FIG. 4A

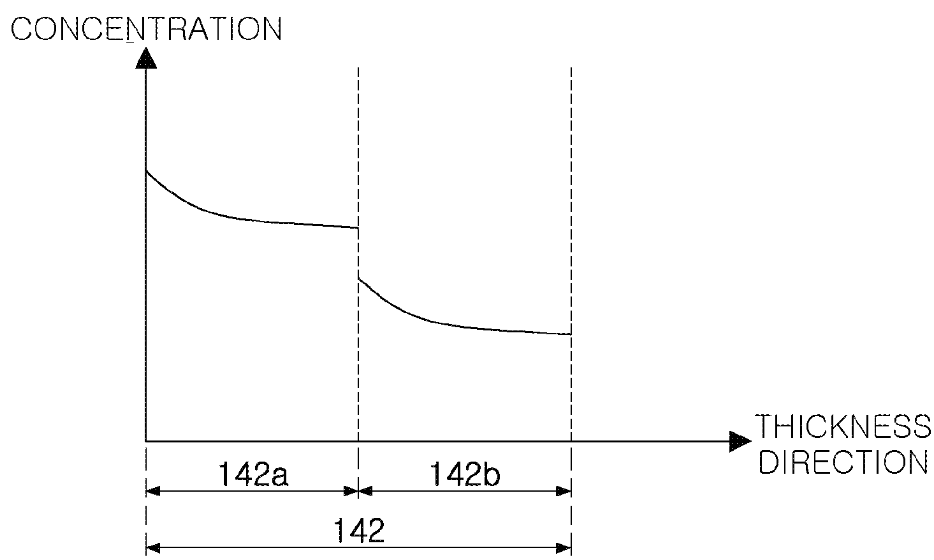

CONCENTRATION
THICKNESS DIRECTION
142a
142b
142

FIG. 4B

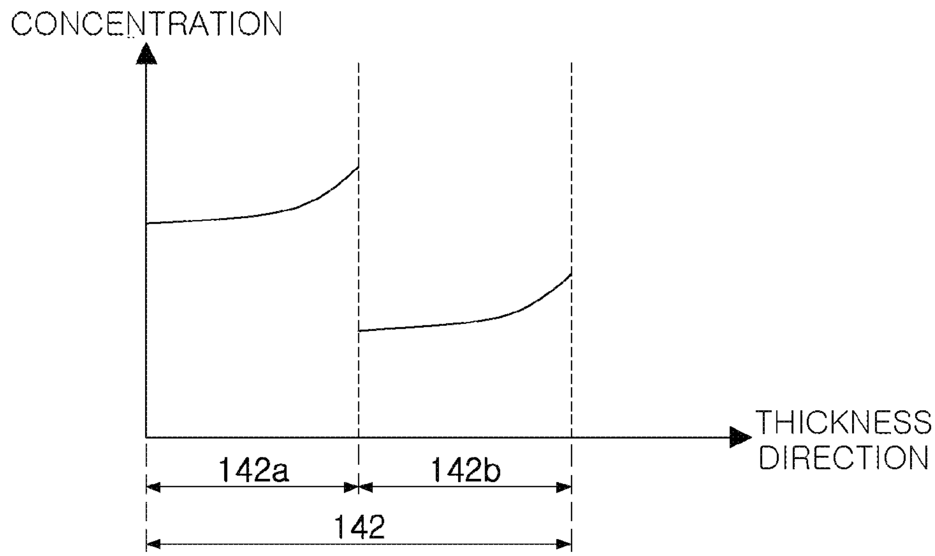

CONCENTRATION
THICKNESS DIRECTION
142a
142b
142

FIG. 4C

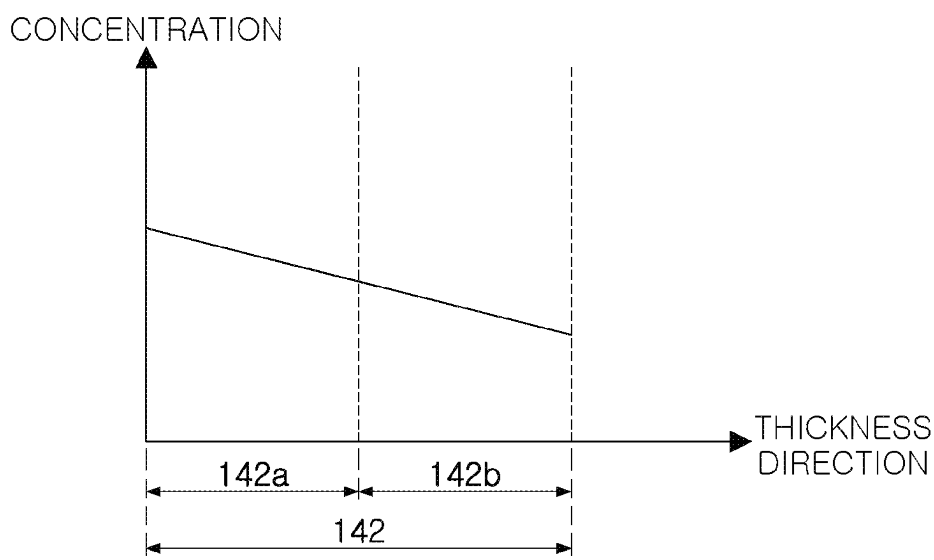

CONCENTRATION
THICKNESS DIRECTION
142a
142b
142

FIG. 4D

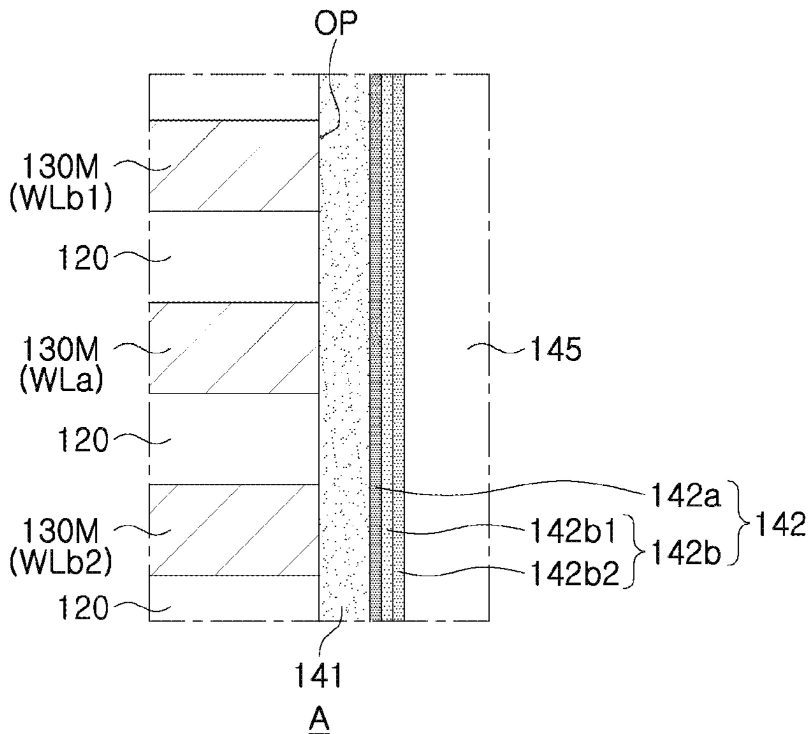

OP
130M
(WLb1)
120
130M
(WLa)
120
130M
(WLb2)
120
145
142a
142b1
142b2
142b
142
141
A

FIG. 5A

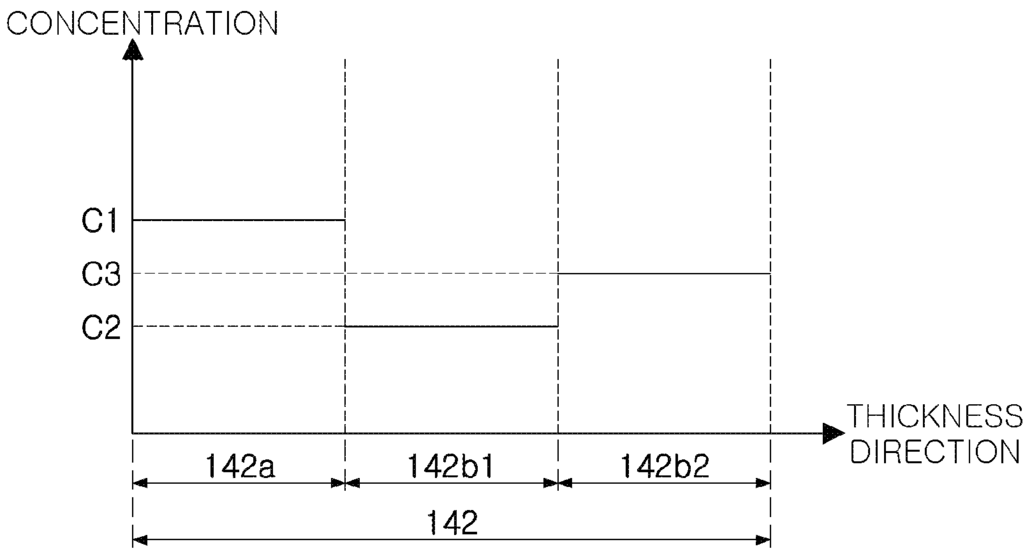

CONCENTRATION
C1
C3
C2
THICKNESS
DIRECTION
142a
142b1
142b2
142

FIG. 5B

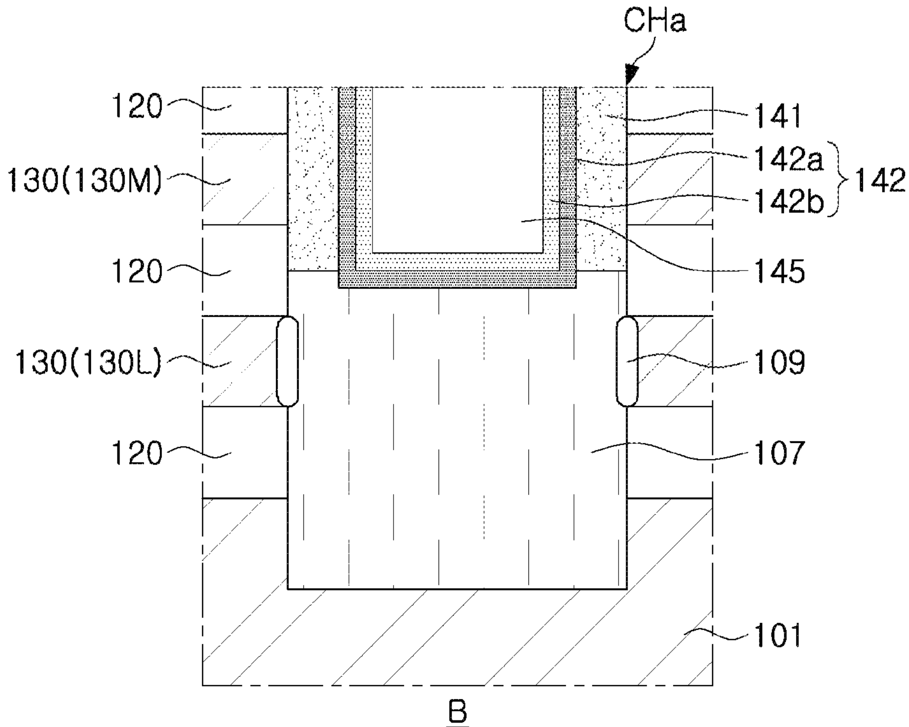

CHa
120
130(130M)
120
130(130L)
120
141
142a
142b
142
145
109
107
101
B

FIG. 6A

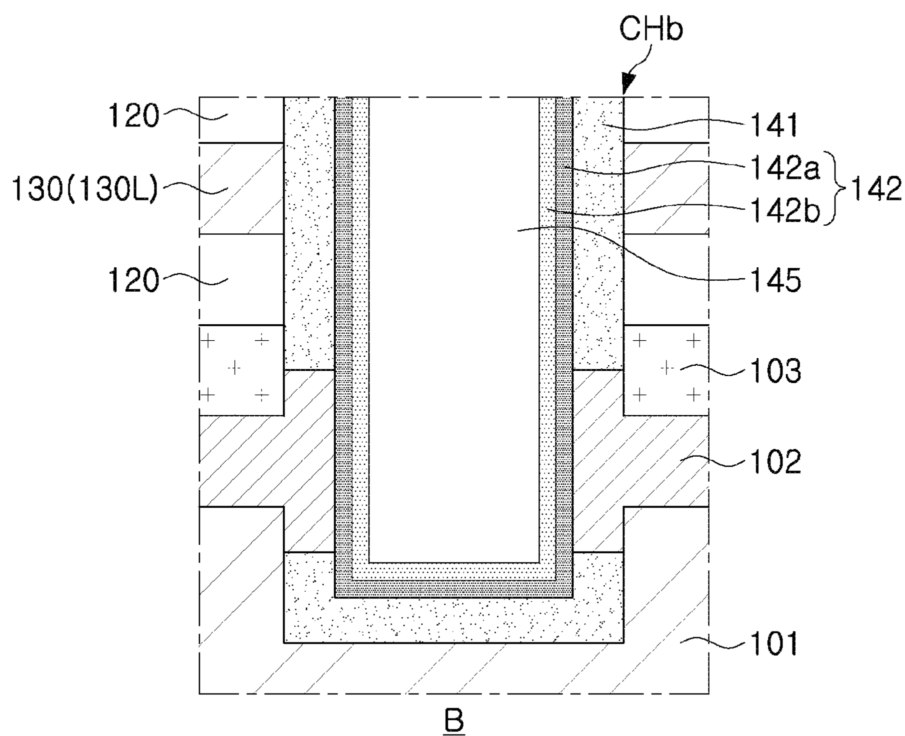

CHb
120
130(130L)
120
141
142a
142b
142
145
103
102
101
B

FIG. 6B

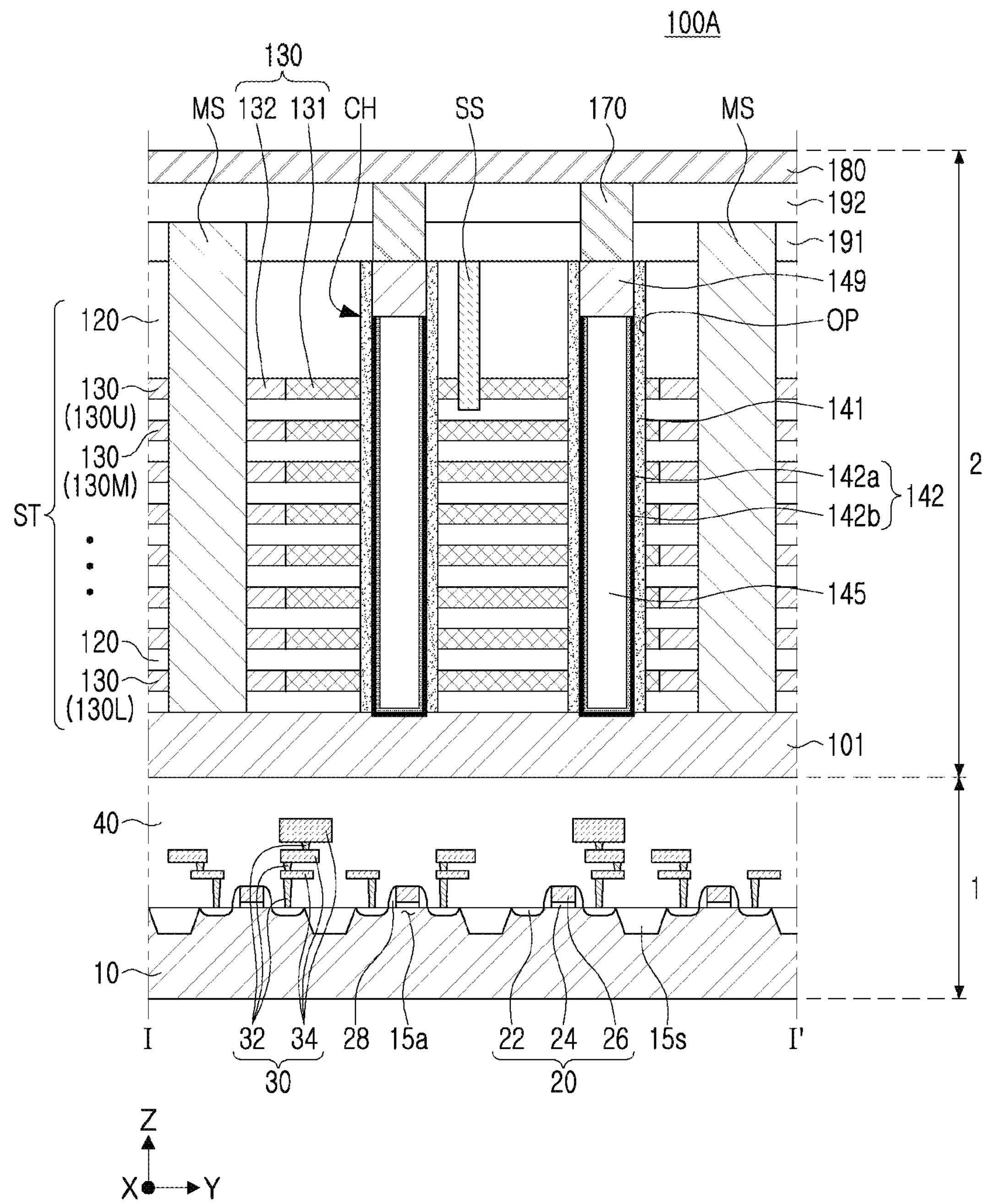
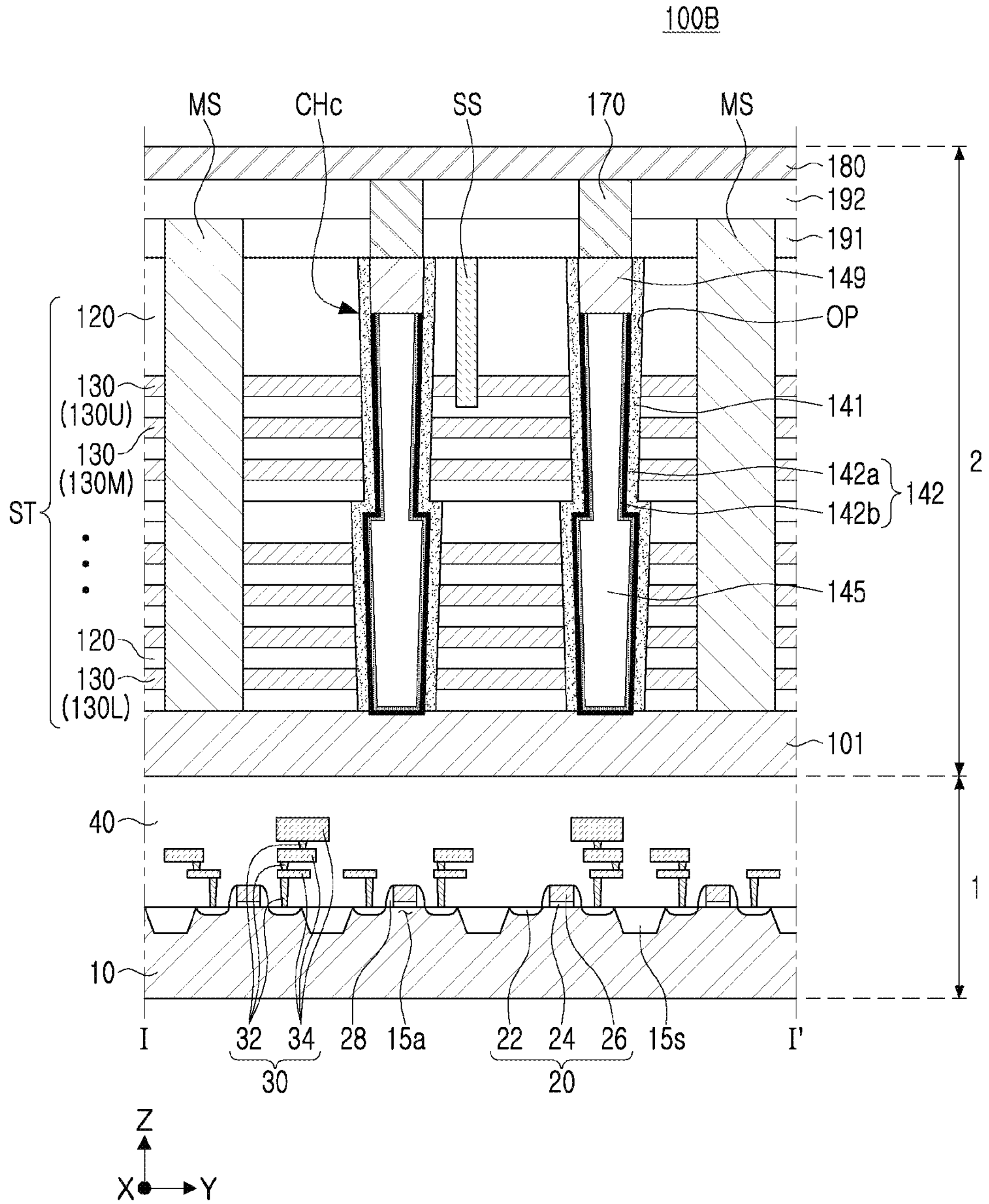

100A
130
MS
132
131
CH
SS
170
MS
180
192
191
149
OP
120
130
(130U)
130
(130M)
ST
141
142a
142b
142
145
2
120
130
(130L)
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 7



100B
MS
CHc
SS
170
MS
180
192
191
149
OP
141
142a
142b
142
145
101
2
1
120
130
(130U)
130
(130M)
ST
120
130
(130L)
40
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 8

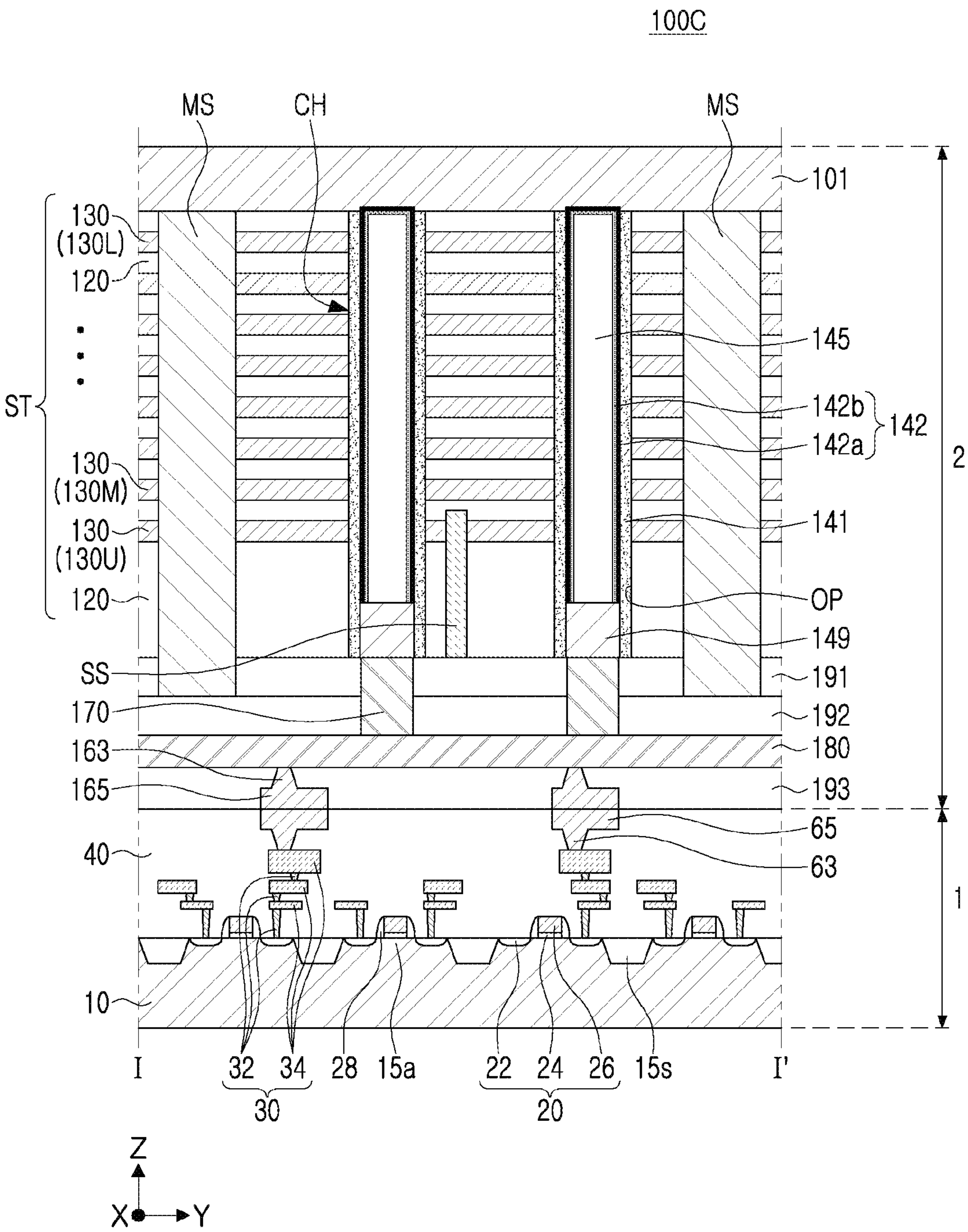

100C
MS
CH
MS
101
130
(130L)
120
ST
130
(130M)
130
(130U)
120
145
142b
142a
142
141
OP
149
2
SS
191
170
192
180
163
193
165
65
63
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 9

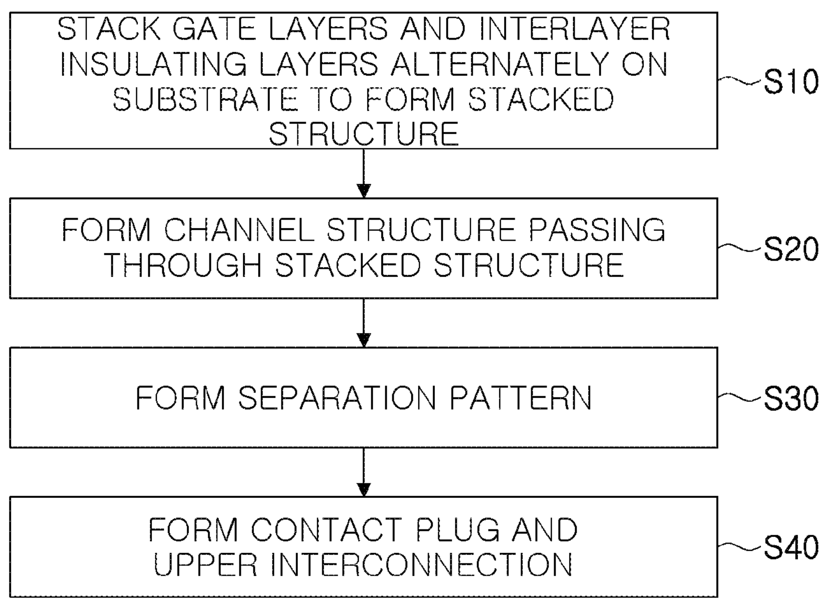

STACK GATE LAYERS AND INTERLAYER INSULATING LAYERS ALTERNATELY ON SUBSTRATE TO FORM STACKED STRUCTURE
S10
FORM CHANNEL STRUCTURE PASSING THROUGH STACKED STRUCTURE
S20
FORM SEPARATION PATTERN
S30
FORM CONTACT PLUG AND UPPER INTERCONNECTION
S40

FIG. 10A

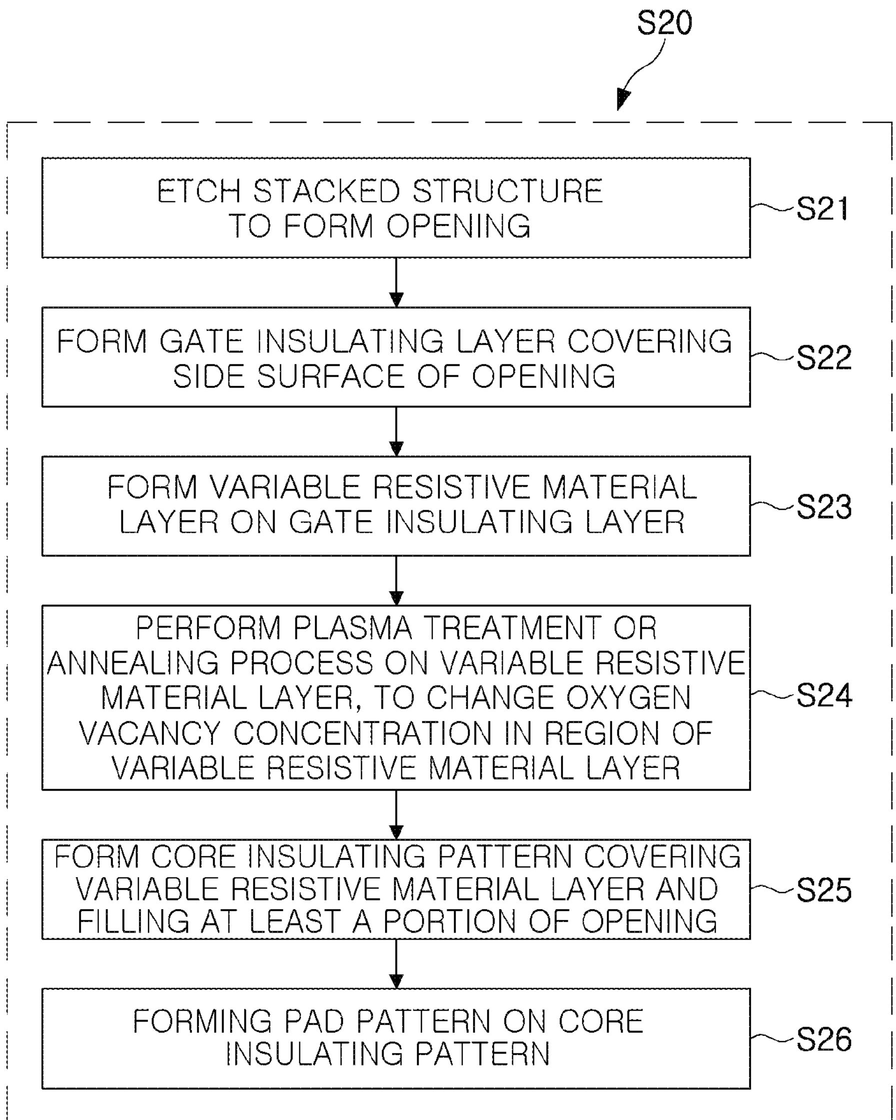

S20
ETCH STACKED STRUCTURE TO FORM OPENING
S21
FORM GATE INSULATING LAYER COVERING SIDE SURFACE OF OPENING
S22
FORM VARIABLE RESISTIVE MATERIAL LAYER ON GATE INSULATING LAYER
S23
PERFORM PLASMA TREATMENT OR ANNEALING PROCESS ON VARIABLE RESISTIVE MATERIAL LAYER, TO CHANGE OXYGEN VACANCY CONCENTRATION IN REGION OF VARIABLE RESISTIVE MATERIAL LAYER
S24
FORM CORE INSULATING PATTERN COVERING VARIABLE RESISTIVE MATERIAL LAYER AND FILLING AT LEAST A PORTION OF OPENING
S25
FORMING PAD PATTERN ON CORE INSULATING PATTERN
S26

FIG. 10B

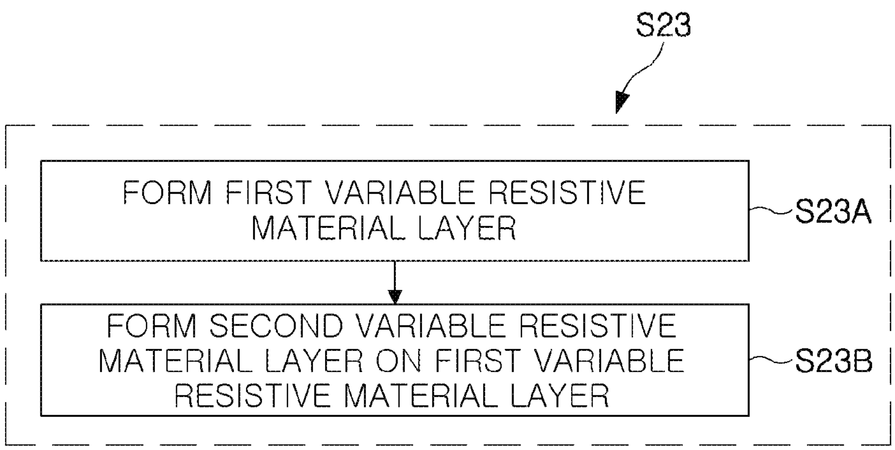


FIG. 11A

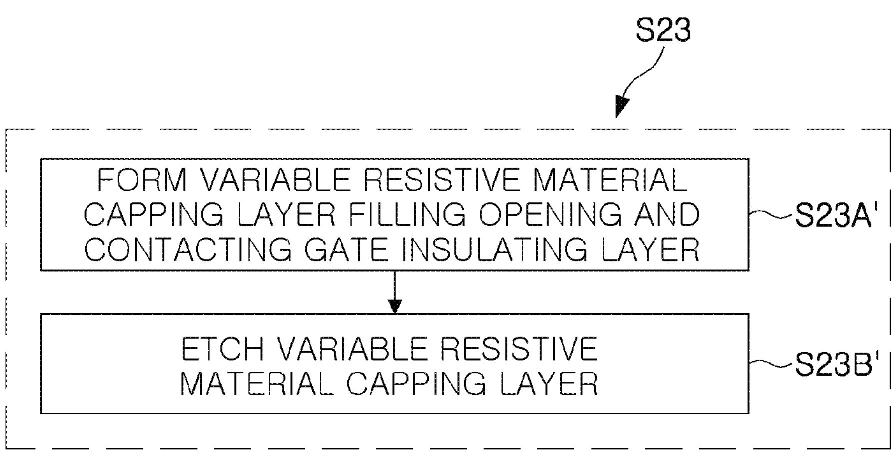


FIG. 11B

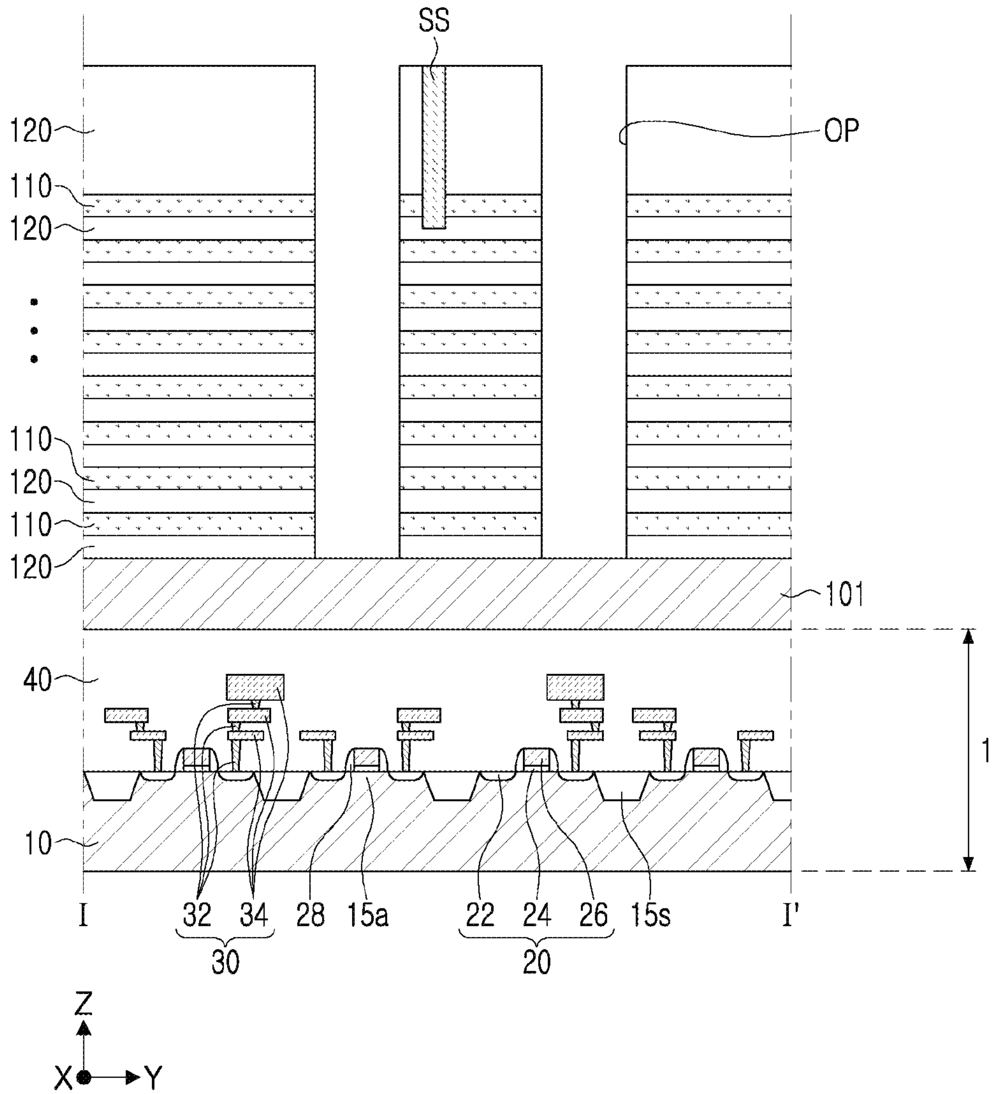

SS
120
OP
110
120
110
120
110
120
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 12

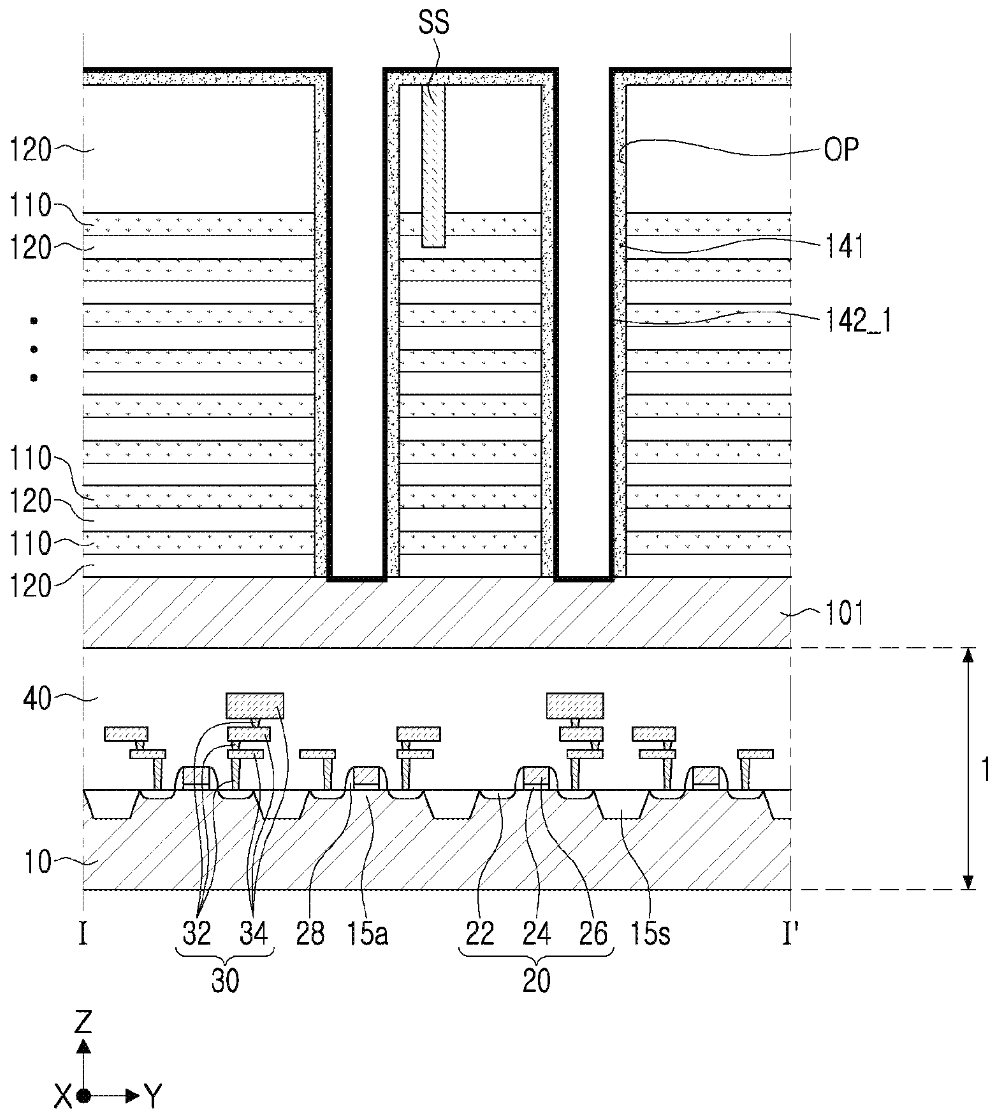

SS
OP
120
110
120
141
142_1
110
120
110
120
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 13A

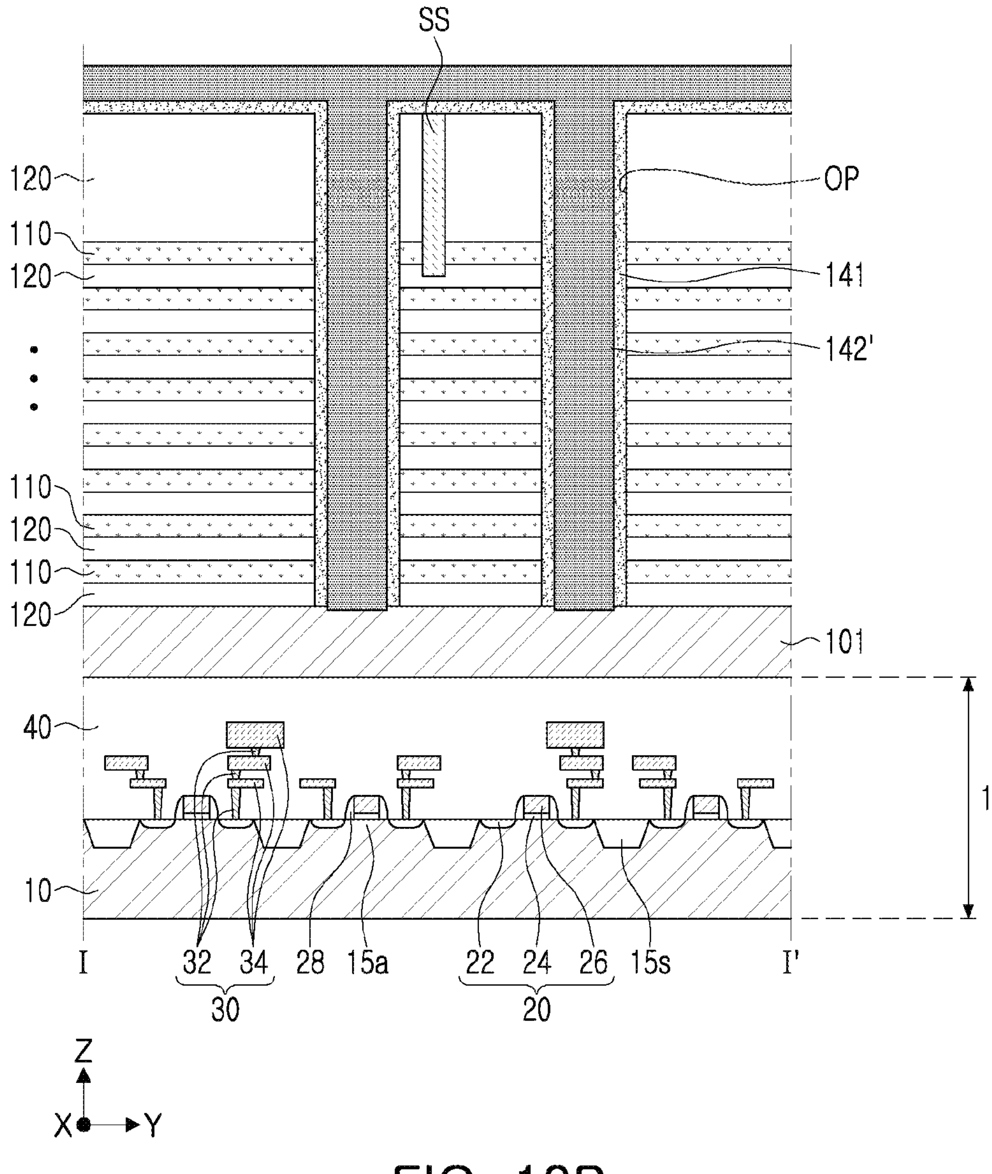

SS
120
110
120
OP
141
142'
110
120
110
120
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 13B

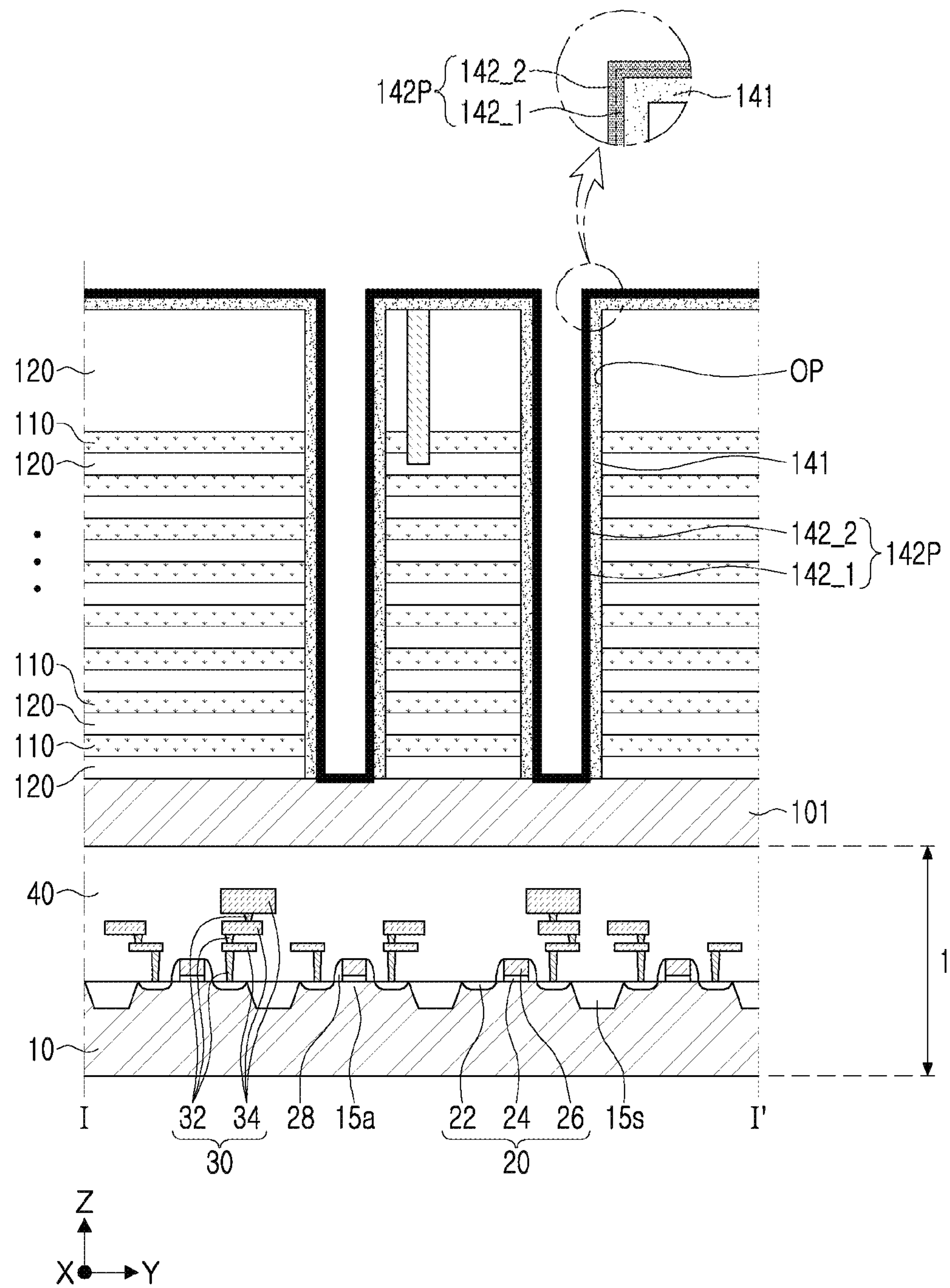

142P
142_2
142_1
141
120
110
120
OP
141
142_2
142_1
142P
110
120
110
120
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 14

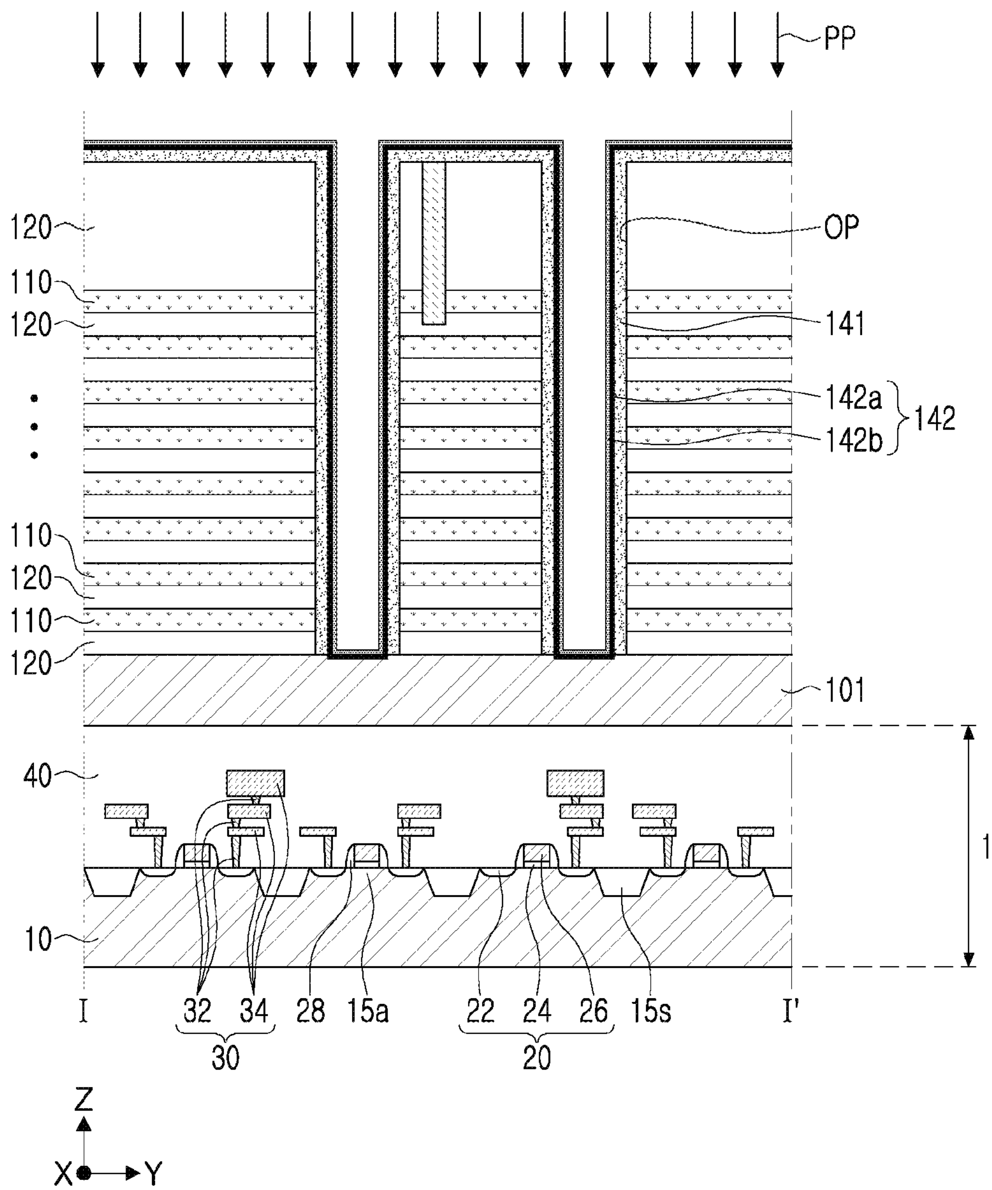
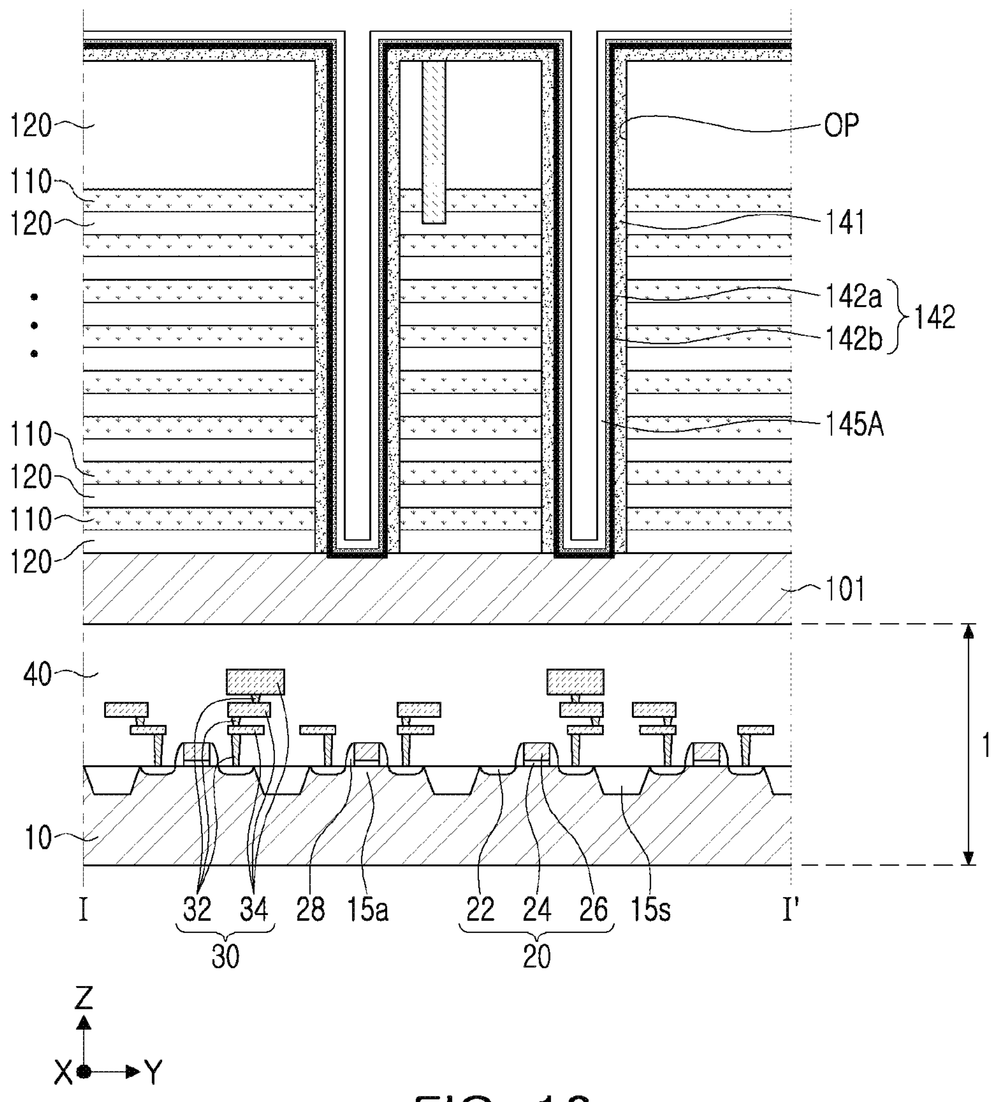

PP
120
110
120
OP
141
142a
142
142b
110
120
110
120
101
40
1
10
I
32
34
30
28
15a
22
24
26
20
15s
I'
Z
X
Y

FIG. 15



120
110
120
110
120
110
120
OP
141
142a
142b
142
145A
101
40
1
10
I
32
34
30
28
15a
22
24
26
20
15s
I'
Z
X
Y

FIG. 16

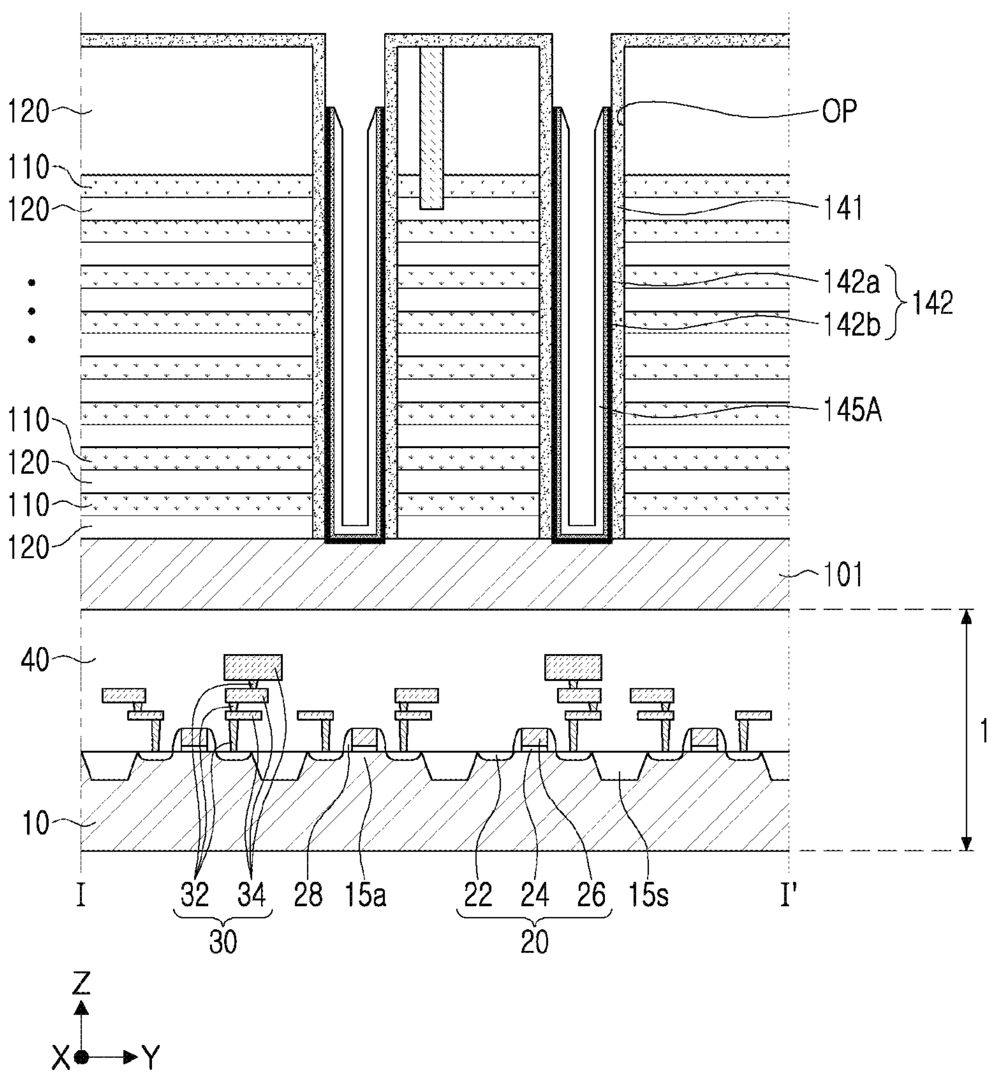

120
110
120
110
120
110
120
OP
141
142a
142b
142
145A
101
40
1
10
I
32
34
30
28
15a
22
24
26
20
15s
I'
Z
X
Y

FIG. 17

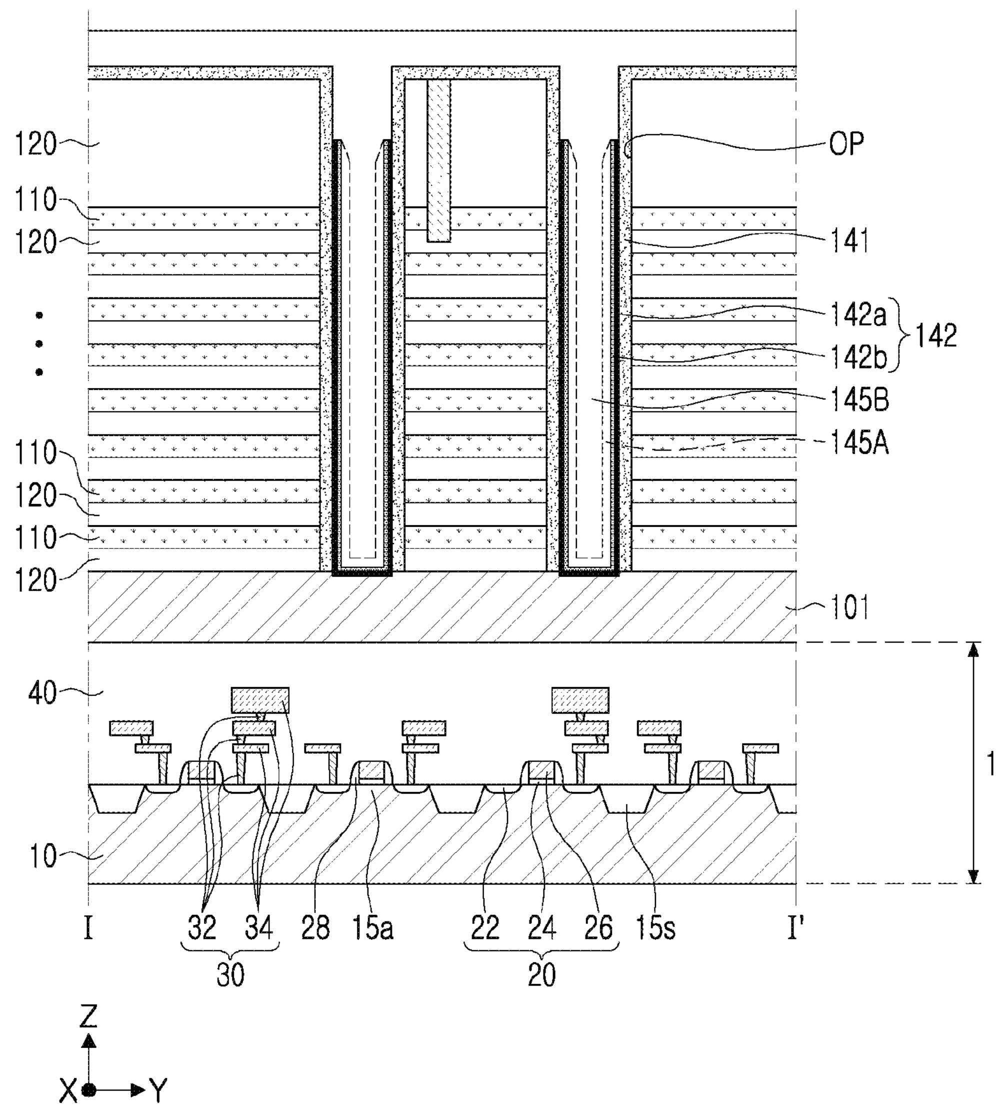
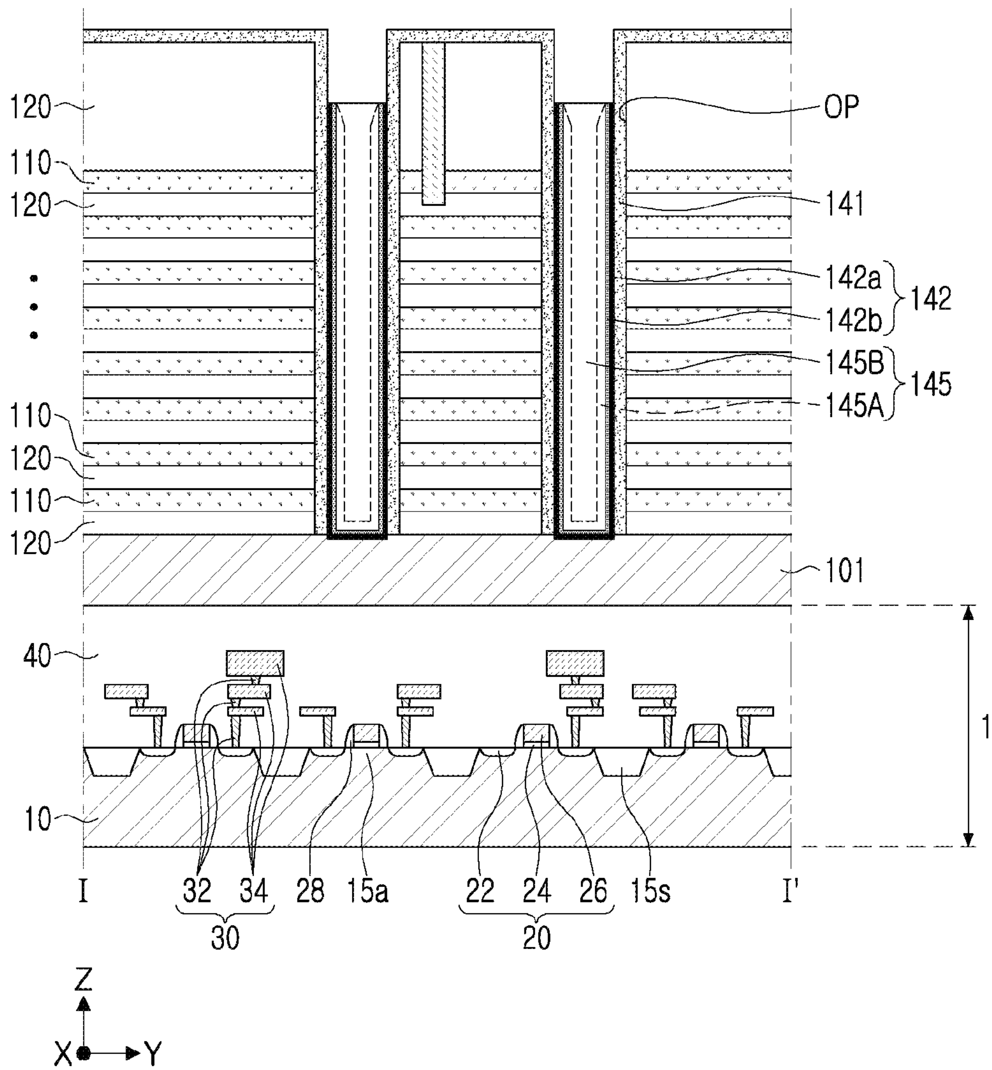

120
110
120
110
120
110
120
OP
141
142a
142b
142
145B
145A
101
40
1
10
I
32
34
30
28
15a
22
24
26
20
15s
I'
Z
X
Y

FIG. 18



120
110
120
110
120
110
120
OP
141
142a
142b
142
145B
145A
145
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 19

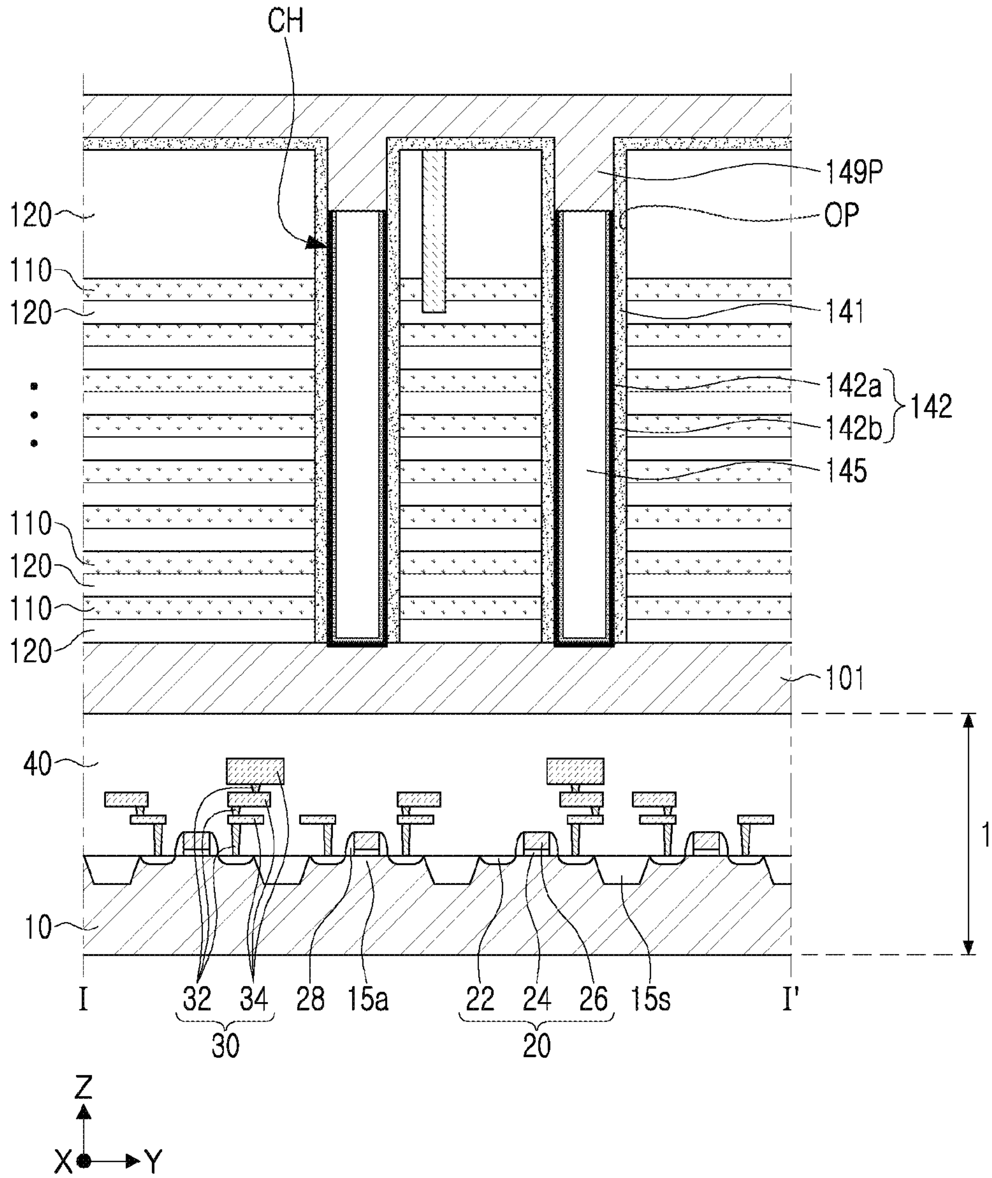

CH
149P
OP
120
110
120
141
142a
142
142b
145
110
120
110
120
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 20

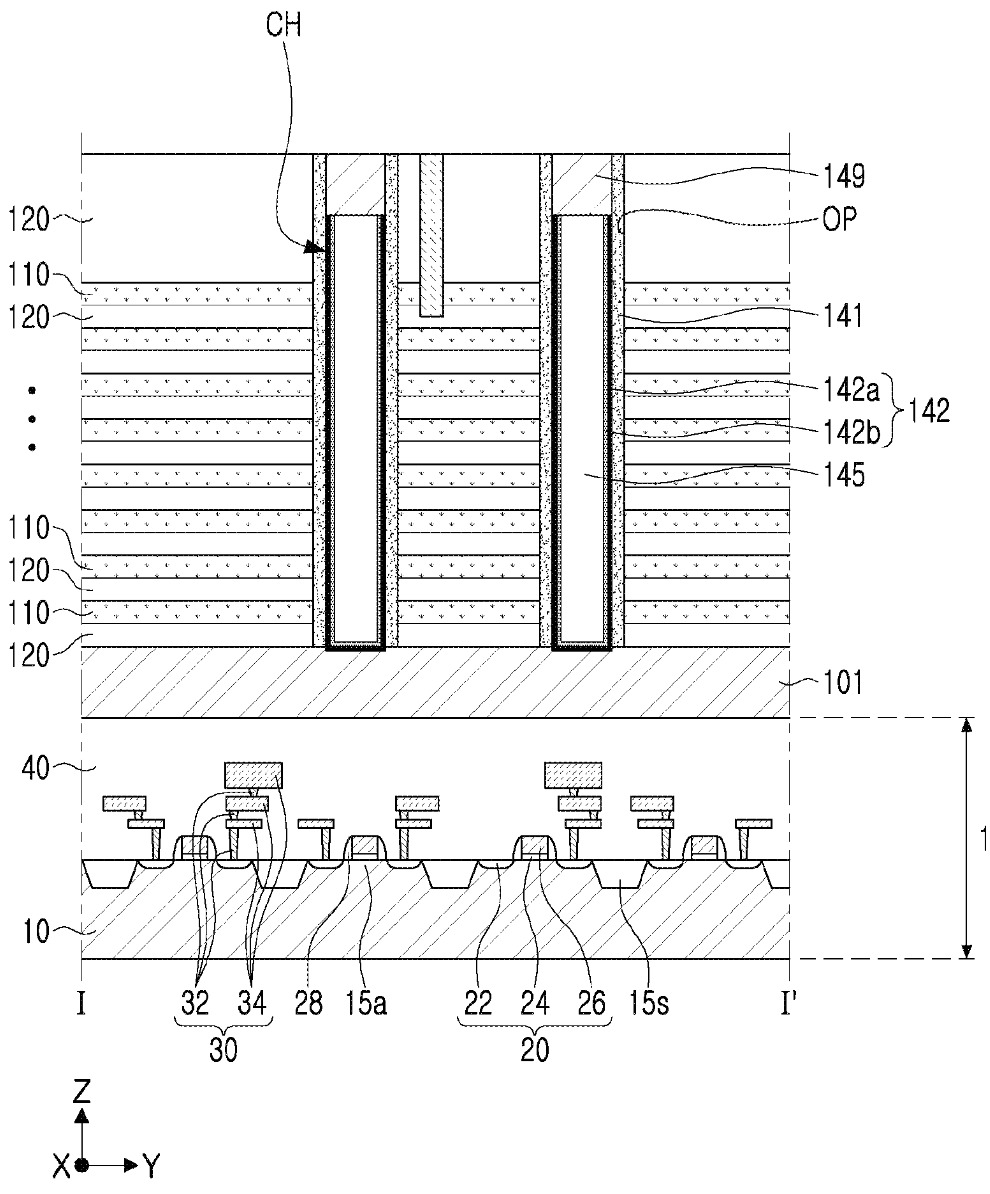

CH
149
OP
120
110
120
141
142a
142
142b
145
110
120
110
120
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 21

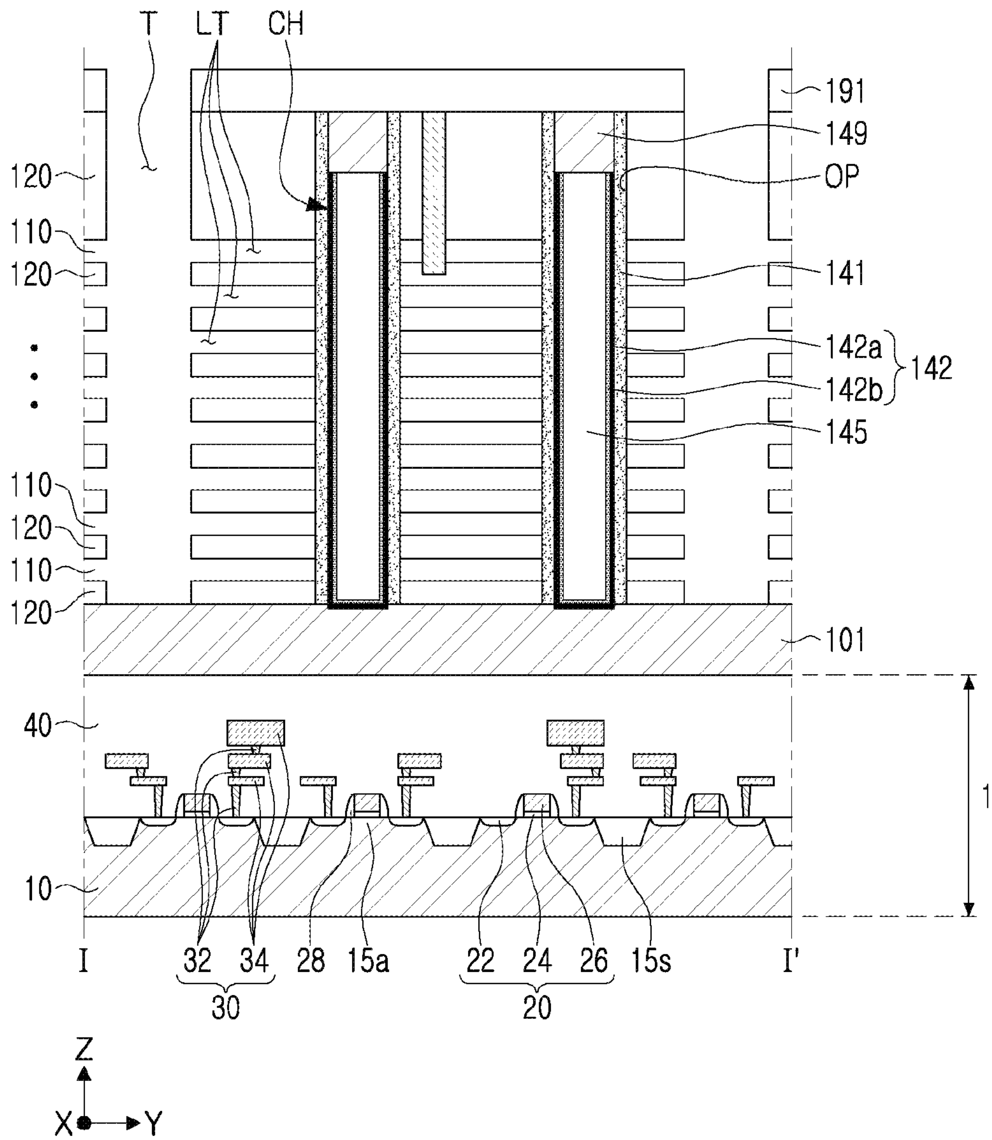

T
LT
CH
191
149
OP
120
110
141
142a
142b
142
145
101
40
1
10
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 22

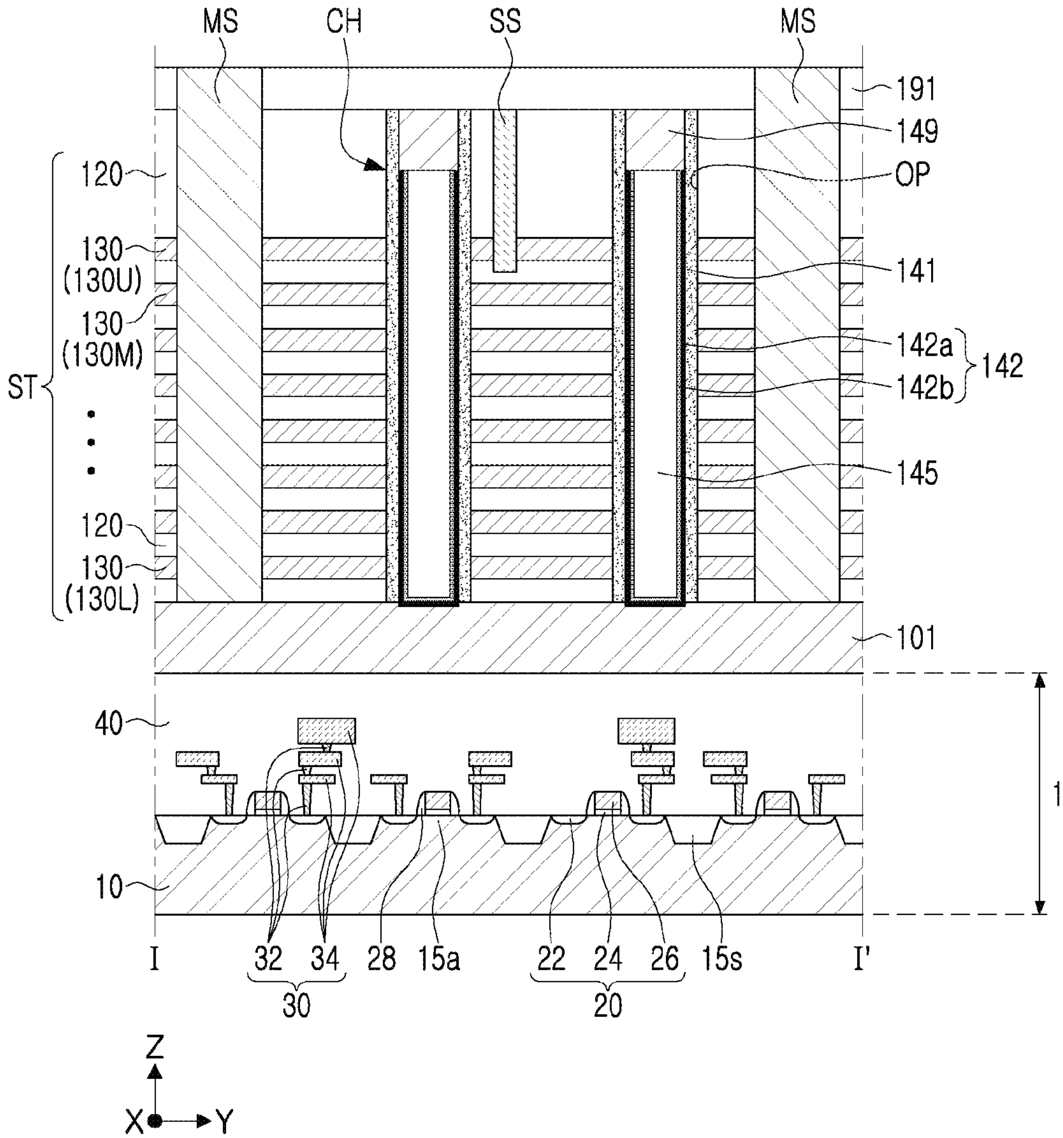

MS
CH
SS
MS
191
149
OP
141
142a
142b
142
145
101
120
130
(130U)
130
(130M)
ST
120
130
(130L)
40
10
1
I
32
34
28
15a
22
24
26
15s
I'
30
20
Z
X
Y

FIG. 23

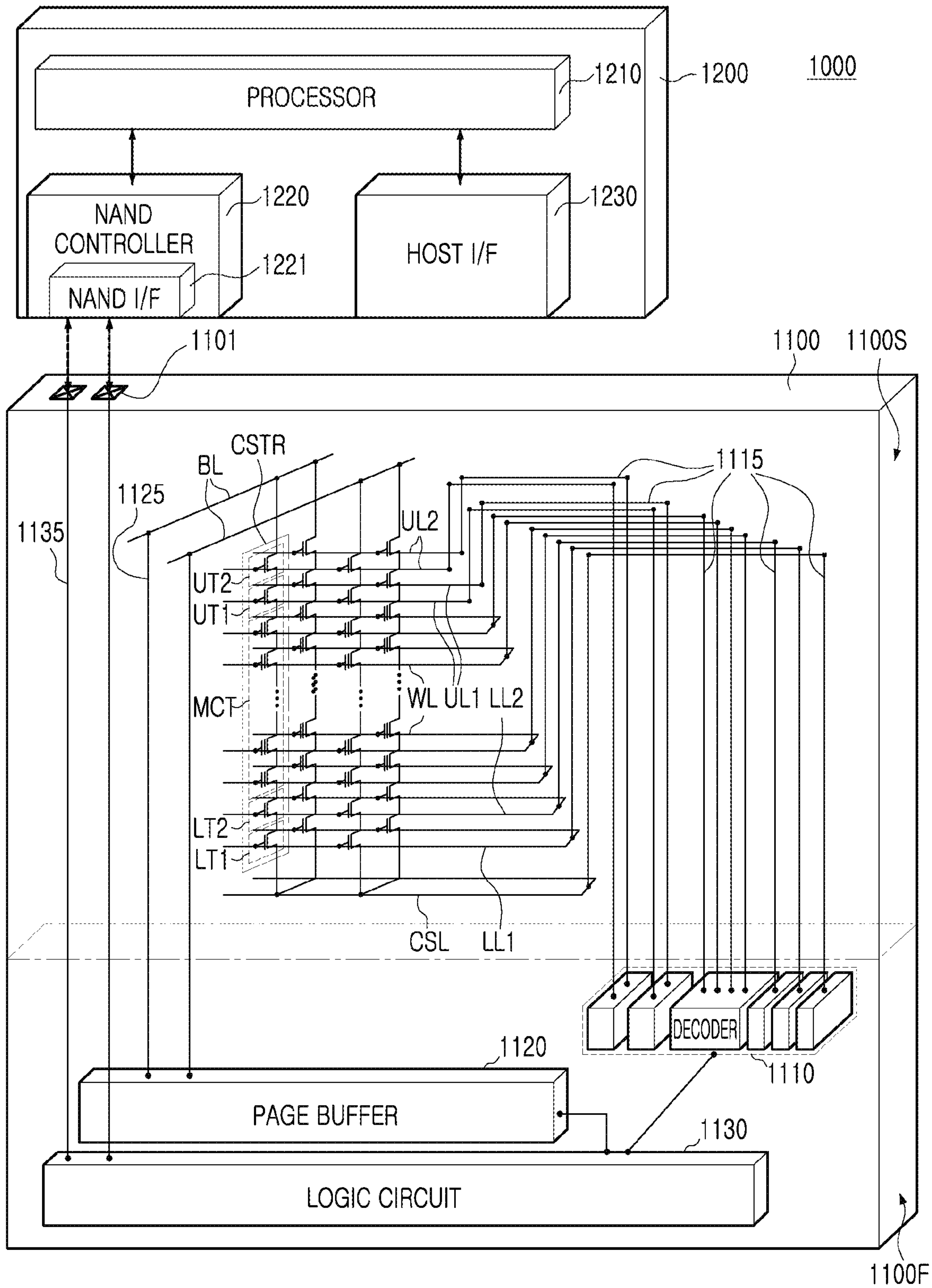
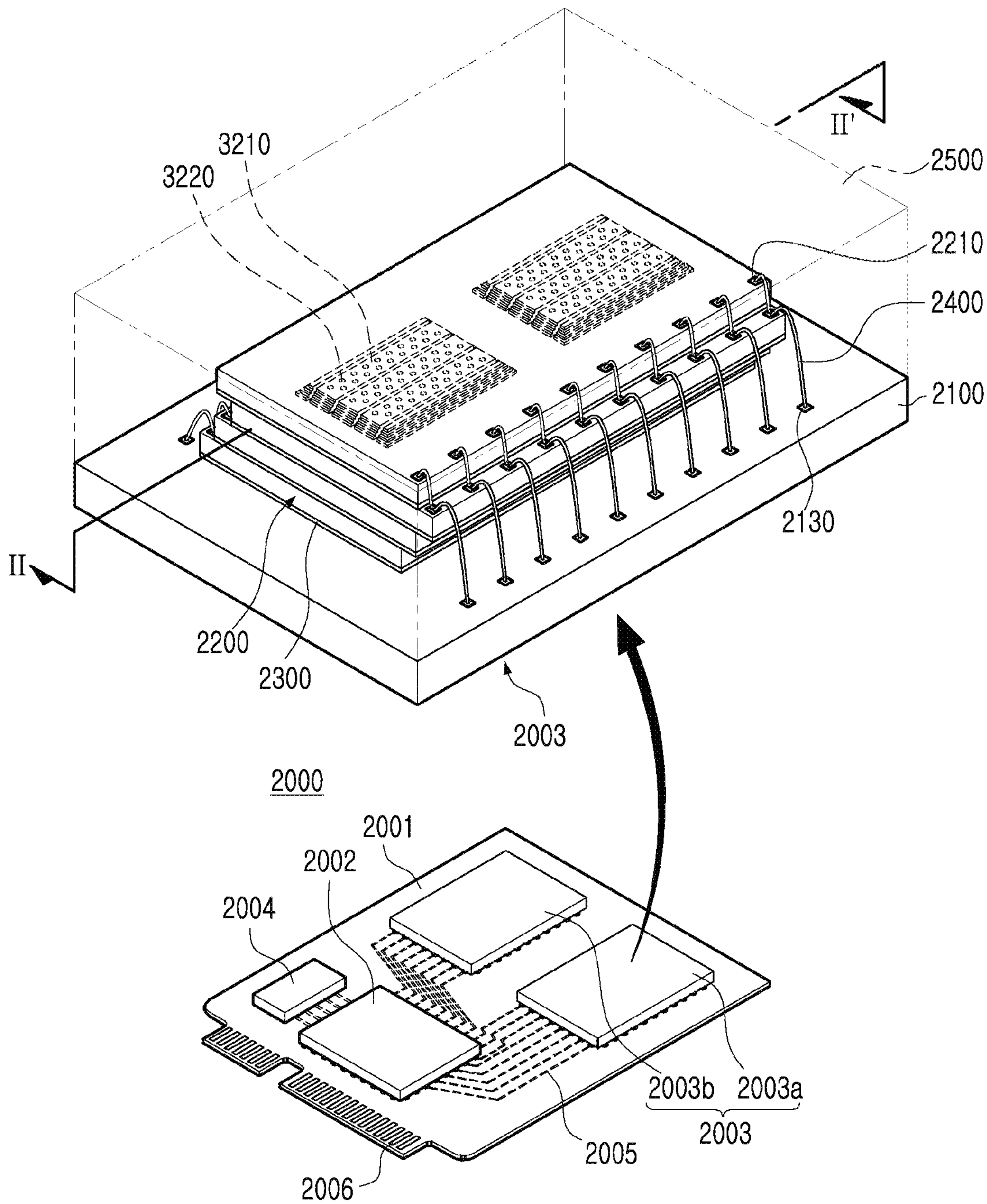

1000
PROCESSOR
1210
1200
NAND CONTROLLER
1220
NAND I/F
1221
HOST I/F
1230
1101
1100
1100S
1135
1125
BL
CSTR
1115
UL2
UT2
UT1
MCT
WL UL1 LL2
LT2
LT1
CSL
LL1
DECODER
1110
1120
PAGE BUFFER
1130
LOGIC CIRCUIT
1100F

FIG. 24



3210
3220
II'
2500
2210
2400
2100
2130
II
2200
2300
2003
2000
2001
2002
2004
2003b
2003a
2003
2005
2006

FIG. 25

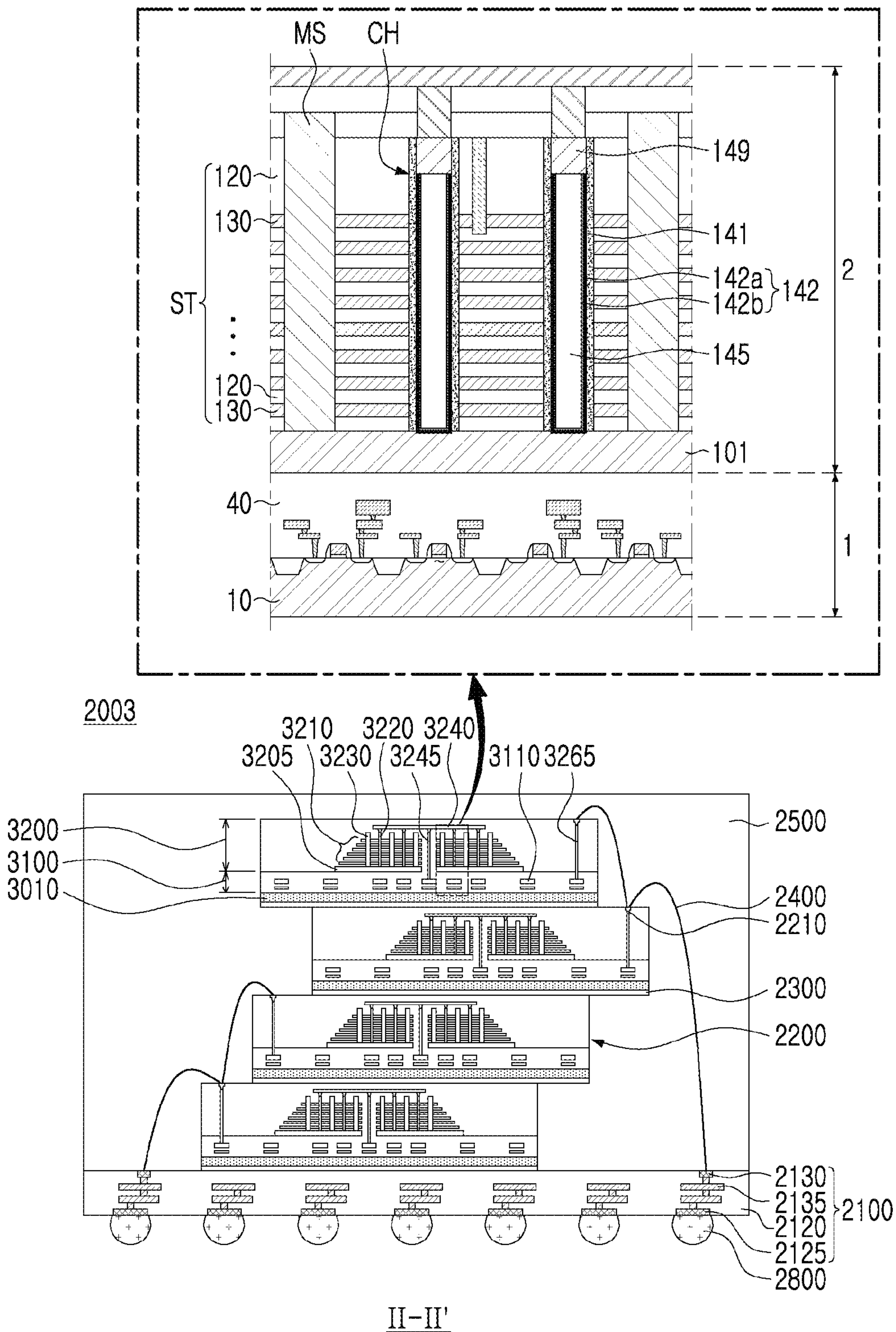

MS
CH
149
120
130
141
142a
142b
142
2
ST
145
120
130
101
40
1
10
2003
3210
3220
3240
3205
3230
3245
3110
3265
3200
3100
3010
2500
2400
2210
2300
2200
2130
2135
2120
2125
2100
2800
II-II'

FIG. 26

METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims benefit of priority to Korean Patent Application No. 10-2022-0000680 filed on Jan. 4, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

Some example embodiments relate to a method of manufacturing or fabricating a semiconductor device and/or a semiconductor device.

In a data storage system requiring data storage, a semiconductor device capable of storing high-capacity data is required or desired. Accordingly, a method of increasing data storage capacity of a semiconductor device is being researched. For example, as a method for increasing the data storage capacity of a semiconductor device, a semiconductor device including three-dimensionally arranged memory cells, instead of two-dimensionally arranged memory cells, has been proposed.

SUMMARY

Some example embodiments may provide a method of manufacturing a semiconductor device having improved electrical characteristics and/or a simplified manufacturing process.

Alternatively or additionally, some example embodiments may provide a semiconductor device having improved electrical characteristics and a simplified manufacturing process, and/or a data storage system including the same.

According some example embodiments, a method of manufacturing a semiconductor device includes forming a stacked structure by stacking gate layers and interlayer insulating layers alternately on a substrate; and forming a channel structure passing through the stacked structure in a vertical direction, wherein the forming a channel structure includes forming an opening by etching the stacked structure; forming a gate insulating layer covering at least a side surface of the opening; forming a variable resistive material layer on the gate insulating layer; changing an oxygen vacancy concentration in a region of the variable resistive material layer by performing one or both of a plasma treatment process or an annealing process on the variable resistive material layer; forming a core insulating pattern that covers the variable resistive material layer and fills at least a portion of the opening, after the performing the one or both of plasma treatment process and the annealing process; and forming a pad pattern on the core insulating pattern.

According to some example embodiments, a method of manufacturing a semiconductor device includes forming a stacked structure by stacking gate layers and interlayer insulating layers alternately on a substrate; and forming a channel structure passing through the stacked structure in a vertical direction. The forming a channel structure includes forming an opening by etching the stacked structure; forming a gate insulating layer covering at least a side surface of the opening; forming a variable resistive material layer on the gate insulating layer and including a first region and a second region; changing an oxygen vacancy concentration in a first subregion of the first region or a second subregion of the second region; forming a core insulating pattern filling at least a portion of the opening; and forming a pad pattern on the core insulating pattern.

According to some example embodiments, a method of manufacturing a semiconductor device includes forming a stacked structure by stacking gate layers and interlayer insulating layers alternately on a substrate; and forming a channel structure passing through the stacked structure in a vertical direction. The forming a channel structure includes forming an opening by etching the stacked structure; forming a gate insulating layer in the opening; forming a variable resistive material capping layer filling the opening and contacting the gate insulating layer; etching a central region of the variable resistive material capping layer to form a variable resistive material layer extending along a side surface of the gate insulation layer and having a specific thickness from the side surface of the gate insulation layer; changing an oxygen vacancy concentration in a region of the variable resistive material layer by performing one or both of a plasma treatment process or an annealing process on the variable resistive material layer; forming a core insulating pattern that covers the variable resistive material layer and that fills at least a portion of the opening, after performing the one or both of the plasma treatment process or the annealing process; and forming a pad pattern on the core insulating pattern.

According to some example embodiments, a semiconductor device includes a substrate; gate electrodes stacked on the substrate and spaced apart from each other in a vertical direction; and a channel structure in an opening that passes through the gate electrodes in the vertical direction. The channel structure includes a core insulating pattern spaced apart from a side surface of the opening, a gate insulating layer contacting the gate electrodes at the side surface of the opening, and a variable resistive material layer between the gate insulating layer and the core insulating pattern. The variable resistive material layer includes a channel region including oxygen vacancies at a first concentration, and an data storage region including oxygen vacancies at a second concentration, less than the first concentration, wherein the channel region is in contact with the gate insulating layer and extends along a side surface of the gate insulating layer, and the data storage region is in contact with the core insulating pattern and extends along a side surface of the core insulating pattern.

According to some example embodiments, a data storage system includes a semiconductor storage device including a lower substrate, a lower structure including circuit elements on the lower substrate, an upper structure on the lower structure, an input/output pad electrically connected to the circuit elements; and a controller electrically connected to the semiconductor storage device through the input/output pad and configured to control the semiconductor storage device, wherein the semiconductor storage device includes an upper substrate; gate electrodes stacked on the upper substrate spaced apart from each other in a vertical direction; and a channel structure in an opening that passes through the gate electrodes in the vertical direction. The channel structure includes a core insulating pattern spaced apart from a side surface of the opening, a gate insulating layer contacting the gate electrodes at the side surface of the opening, and a variable resistive material layer between the gate insulating layer and the core insulating pattern. The variable resistive material layer includes a channel region including oxygen vacancies at a first concentration, and an data storage region including oxygen vacancies at a second concentration, less than the first concentration, wherein the channel region is in contact with the gate insulating layer and extends along a side surface of the gate insulating layer, and the data storage region is in contact with the core insulating pattern and extends along a side surface of the core insulating pattern.

BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects, features, and advantages of the present inventive concept will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

FIG. **1** is a schematic plan view of a semiconductor device according to various example embodiments.

FIG. **2** is a schematic cross-sectional view of a semiconductor device according to various example embodiments.

FIG. **3** is a partially enlarged view of a portion of a semiconductor device according to various example embodiments.

FIGS. **4**A to **4**D are graphs illustrating an oxygen vacancy concentration in a variable resistive material layer of a semiconductor device according to various example embodiments.

FIG. **5**A is a partially enlarged view of a portion of a semiconductor device according to various example embodiments.

FIG. **5**B is a graph illustrating an oxygen vacancy concentration in a variable resistive material layer of a semiconductor device according to various example embodiments.

FIGS. **6**A and **6**B are partially enlarged views illustrating a portion of a semiconductor device according to various example embodiments.

FIGS. **7** to **9** are schematic cross-sectional views of a semiconductor device according to various example embodiments.

FIGS. **10**A to **10**B, and FIGS. **11**A to **11**B are flowcharts illustrating a method of manufacturing a semiconductor device according to various example embodiments.

FIGS. **12**, FIGS. **13**A to **13**B, and FIGS. **14-23** are schematic views illustrating a method of manufacturing a semiconductor device according to various example embodiments.

FIG. **24** is a view schematically illustrating a data storage system including a semiconductor device according to various example embodiments.

FIG. **25** is a perspective view schematically illustrating a data storage system including a semiconductor device according to various example embodiments.

FIG. **26** is a cross-sectional view schematically illustrating a semiconductor package according to various example embodiments.

DETAILED DESCRIPTION

Hereinafter, various example embodiments of inventive concepts will be described with reference to the accompanying drawings.

FIG. **1** is a schematic plan view of a semiconductor device according to various example embodiments.

FIG. **2** is a schematic cross-sectional view of a semiconductor device according to various example embodiments. FIG. **2** illustrates a cross-sectional view taken along line I-I' of FIG. **1**.

FIG. **3** is a partially enlarged view of a portion of a semiconductor device according to various example embodiments. FIG. **3** illustrates an enlarged view of portion 'A' of FIG. **2**.

Referring to FIGS. **1** to **3**, a semiconductor device **100** may include a first structure **1** including a lower substrate **10** and a second structure **2** including an upper substrate **101**. The second structure **2** may be disposed on the first structure **1**. The first structure **1** may be or correspond to a region in which a peripheral circuit region of the semiconductor device **100** is disposed, and the peripheral circuit region may include one or more of a column decoder, a row decoder, a page buffer, other peripheral circuits, a redundancy circuit, and the like. The second structure **2** may be a region in which memory cells of the semiconductor device **100** are disposed, and may include gate electrodes **130**, channel structures CH, and the like. The semiconductor device **100** may be or may include a cell-over-peri (COP) semiconductor device; however, example embodiments are not limited thereto.

The first structure **1** may include a lower substrate **10**, device isolation layers **15***s* defining an active region **15***a* on the lower substrate **10**, circuit elements **20** disposed on the lower substrate **10**, a lower interconnection structure **30** electrically connected to the circuit elements **20**, and a lower insulating layer **40**.

The lower substrate **10** may include a semiconductor material, for example, one or more of a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI compound semiconductor. The lower substrate **10** may be provided as a bulk wafer or an epitaxial layer. The lower substrate **10** may be doped, e.g. may be lightly doped with one or more of boron, arsenic, or phosphorus; however, example embodiments are not limited thereto. The lower substrate **10** may be disposed below an upper substrate **101**. The device isolation layers **15***s* may be disposed in the lower substrate **10**, and source/drain regions **22** including impurities such as one or more of boron, phosphorus, or arsenic may be disposed in a portion of the active region **15***a*.

The circuit elements **20** may each include a transistor such as a planar transistor including a source/drain region **22**, a circuit gate dielectric layer **24**, and a circuit gate electrode **26**. The source/drain regions **22** may be disposed on both sides of the circuit gate electrode **26** in the active region **15***a*. The circuit gate dielectric layer **24** may be disposed between the active region **15***a* and the circuit gate electrode **26**. Spacer layers **28** may be disposed on both sides of the circuit gate electrode **26**. The circuit gate electrode **26** may include, for example, a material layer such as one or more of tungsten (W), titanium (Ti), tantalum (Ta), tungsten nitride (WN), titanium nitride (TiN), tantalum nitride (TaN), polysilicon, or a metal-semiconductor compound.

The lower interconnection structure **30** may be electrically connected to the circuit elements **20**. The lower interconnection structure **30** may include a lower contact **32** and a lower interconnection **34**. A portion of lower contacts **32** may extend in a Z-direction to be connected to the source/drain regions **22**. The lower contact **32** may electrically connect the lower interconnections **34** disposed on different levels to each other. The lower interconnection structure **30** may include a conductive material, for example, a metal material such as one or more of tungsten (W), titanium (Ti), tantalum (Ta), copper (Cu), aluminum (Al), cobalt (Co), molybdenum (Mo), ruthenium (Ru), or the like. A barrier layer formed of a material such as one or more of tungsten nitride (WN), titanium (Ti), titanium nitride (TiN), or the like may be disposed on bottom and side surfaces of the lower interconnection structure **30**. The number of layers and arrangement of the lower contacts **32** and the lower interconnections **34**, constituting the lower interconnection structure **30**, may be variously changed. At least a portion of the lower interconnections **34** may include a pad layer to which a plurality of through-contact plugs extending downwardly from the second structure **2** are directly connected. In some example embodiments, the plurality of through-contact plugs may be disposed to pass through a through-region formed in a stacked structure ST of the second structure **2**.

The lower insulating layer **40** may be disposed to cover the lower substrate **10**, the circuit elements **20**, and the lower interconnection structure **30**. The lower insulating layer **40** may be formed of an insulating material such as silicon oxide and/or silicon nitride. The lower insulating layer **40** may include a plurality of insulating layers. The lower insulating layer **40** may include an etch stop layer formed of silicon nitride.

The second structure **2** may include an upper substrate **101** on the first structure **1**, a stacked structure ST including gate electrodes **130** spaced apart and stacked on the upper substrate **101**, first separation patterns MS passing through the stacked structure ST and separating the gate electrodes **130**, channel structures CH passing through the stacked structure ST, a second separation pattern SS separating upper gate electrodes **130**U among the gate electrodes **130** between the first separation patterns MS, and bit lines **180** disposed on the stacked structure ST. The second structure **2** may further include interlayer insulating layers **120** with which the gate electrodes **130** are alternately stacked and forming a portion of the stacked structure ST, and contact plugs **170** and upper insulating layers **191** and **192**, arranged between the channel structures CH and the bit lines **180**.

The upper substrate **101** may include a semiconductor material, for example, one or more of a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI compound semiconductor. The upper substrate **101** may include, for example, a polysilicon layer having N-type or P-type conductivity. The upper substrate **101** may include an impurity region contacting the channel structure CH.

The gate electrodes **130** may be stacked on the upper substrate **101** to be spaced apart from each other in the Z-direction, and may form a portion of the stacked structure ST. The gate electrodes **130** may extend in the X-direction. The gate electrodes **130** may include a lower gate electrode **130**L forming a gate of a ground select transistor, memory gate electrodes **130**M forming a plurality of memory cells, and upper gate electrodes **130**U forming gates of string select transistors. The number of the memory gate electrodes **130**M constituting the memory cells may be determined according to capacity of the semiconductor device **100**. In some example embodiments, the number of the gate electrodes constituting the string select transistor may be one or two or more, and the number of the gate electrodes constituting the ground select transistor may be one or two or more.

The gate electrodes **130** may be vertically spaced apart and stacked on the upper substrate **101**, and although not illustrated, may extend by different lengths in a Y-direction to form a stepped structure or a stair structure. The gate electrodes **130** may have pad regions in which a lower gate electrode among the gate electrodes **130** is extended to be longer than an upper gate electrode among the gate electrodes **130** due to the stepped structure. Gate contact plugs may be connected to the gate electrodes **130** through the pad regions of the gate electrodes **130**. In some example embodiments, the gate contact plugs may be electrically connected to the circuit elements **20** of the first structure **1** through through-contact plugs passing through a through-region disposed in the stacked structure ST.

The gate electrodes **130** may be arranged to be separated from each other in the Y-direction by the first separation patterns MS extending in an X-direction. The gate electrodes **130** between a pair of first separation patterns MS may form one memory block, but a scope of the memory block is not limited thereto. The gate electrodes **130** may include a first layer and a second layer, respectively. The first layer may cover upper and lower surfaces of the second layer, and may extend between the channel structures CH and the second layer. The first layer may include a high-k material such as aluminum oxide (AlO) or the like, and the second layer may include at least one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), tungsten (W), or tungsten nitride (WN). In some example embodiments, the gate electrodes **130** may include polysilicon and/or a metal-semiconductor compound.

The interlayer insulating layers **120** may be disposed between the gate electrodes **130**, and may form the stacked structure ST. Like the gate electrodes **130**, the interlayer insulating layers **120** may be spaced apart from each other in the Z-direction, and may be disposed to extend in the X-direction. The interlayer insulating layers **120** may include an insulating material such as silicon oxide. A portion of the interlayer insulating layers **120** may have different thicknesses. For example, an uppermost interlayer insulating layer **120**, among the interlayer insulating layers **120**, may have a thickness, greater than a thickness of each of the other interlayer insulating layers **120**.

The first separation patterns MS may be disposed to pass through the gate electrodes **130** of the stacked structure ST in the Z-direction, and extend in the X-direction. First separation patterns MS adjacent in the Y-direction may be disposed parallel to each other. The first separation patterns MS may entirely pass through the gate electrodes **130** of the stacked structure ST in the Z-direction, to contact the upper substrate **101**. The first separation patterns MS may be formed of an insulating material, for example, silicon oxide. In some example embodiments, each of the first separation patterns MS may include a core pattern including a conductive material and contacting the upper substrate **101**, and a separation insulation pattern covering a side surface of the core pattern and including an insulating material.

As illustrated in FIG. **1**, the channel structures CH may form a memory cell string, and may be disposed to be spaced apart from each other while forming rows and columns. The channel structures CH may be disposed to form a grid pattern between the first separation patterns MS or may be disposed to form a zigzag shape in a direction. The channel structures CH may be disposed in an opening OP passing through the stacked structure ST in the Z-direction. The channel structures CH may have a pillar shape, and may have inclined side surfaces, narrower in width, as they approach the upper substrate **101** according to an aspect ratio. The channel structures CH may have a tapered pillar profile.

Each of the channel structures CH may include a gate insulating layer **141**, a variable resistive material layer **142**, a core insulating pattern **145**, and a pad pattern **149**. The variable resistive material layer **142** may be formed in an annular shape to cover or surround an outer side surface of the core insulating pattern **145**. The gate insulating layer **141** may be formed in an annular shape to cover or surround an outer side surface of the variable resistive material layer **142**. The gate insulating layer **141**, the variable resistive material layer **142**, and the core insulating pattern **145** may be sequentially disposed from side surfaces of the gate electrodes **130**. For example, the gate insulating layer **141** may be in contact with the gate electrodes **130** on a side surface of the opening OP, the core insulating pattern **145** may be spaced apart from the side surface of the opening OP, and the variable resistive material layer **142** may be disposed between the gate insulating layer **141** and the core insulating pattern **145**.

The gate insulating layer **141** may be disposed between the gate electrodes **130** and the variable resistive material layer **142**. The gate insulating layer **141** may extend along the side surface of the opening OP. The gate insulating layer **141** may extend from a level lower than the lower gate electrode **130**L to a level higher than the upper gate electrode **130**U. An upper surface of the gate insulating layer **141** may be coplanar with or substantially coplanar with an upper surface of the pad pattern **149**. The gate insulating layer **141** may be formed of silicon oxide or doped silicon oxide such as nitrogen-doped silicon oxide.

The variable resistive material layer **142** may cover side and lower surfaces of the core insulating pattern **145**, and may be in contact with the upper substrate **101**. The variable resistive material layer **142** may be formed as a single layer including a transition metal oxide, and may include regions having different oxygen vacancy concentrations. For example, the variable resistive material layer **142** may include a first region **142***a* (corresponding to a ‘channel region’ of a memory cell transistor, and, hereinafter, referred to as a ‘channel region’) having an oxygen vacancy of a first concentration, and a second region **142***b* (corresponding to an ‘data storage region’ of the memory cell transistor, and, hereinafter, referred to as a ‘data storage region’) having an oxygen vacancy of a second concentration, different from the first concentration. The second concentration may be lower than the first concentration. As used herein, an “oxygen vacancy” may refer to a point, such as a point defect, in a unit cell of a crystalline lattice of atoms in an oxide crystal that is absent of an oxygen atom. An oxygen vacancy concentration may indicate a concentration in units of vacancies per volume of unit cells in an oxide crystalline lattice that are void of oxygen.

An oxygen vacancy concentration in the channel region **142***a* of the variable resistive material layer **142** may be greater than an oxygen vacancy concentration in the data storage region **142***b* of the variable resistive material layer **142**. The channel region **142***a* may be in contact with the gate insulating layer **141**, and the data storage region **142***b* may be in contact with the core insulating pattern **145**. In the drawings, a line between the channel region **142***a* and the data storage region **142***b* is illustrated for convenience of explanation, but an interface between the channel region **142***a* and the data storage region **142***b* may not exist. Since the channel region **142***a* and the data storage region **142***b* may be regions formed of the same material, resistance therebetween may be relatively lower than contact resistance due to an interface formed between them, formed of different materials.

A lower portion of the channel region **142***a* of the variable resistive material layer **142** may be in contact with the upper substrate **101**. Since the channel region **142***a* of the variable resistive material layer **142** corresponds to a channel region of the memory cell transistor as described above, a further channel layer may not be disposed between the variable resistive material layer **142** and the gate insulating layer **141**. For example, the variable resistive material layer **142** may be in contact with the gate insulating layer **141** without interposing a polysilicon layer between the variable resistive material layer **142** and the gate insulating layer **141**. For example, the channel region **142***a* of the variable resistive material layer **142** may not include polysilicon. The channel region **142***a* of the variable resistive material layer **142** may be a region in which a portion of the variable resistive material layer **142** is subjected to a plasma treatment process and/or an annealing process to increase an oxygen vacancy concentration. The annealing process may be or may include a thermal annealing process and/or a laser annealing process. Since a portion of the variable resistive material layer **142** may be post-processed to change electrical characteristics, and may be used as the channel region **142***a* of the memory cell transistor, an operation of forming a further channel layer including a material, different from the variable resistive material layer **142**, may be omitted. There may be a reduction in fabrication time, and/or an improvement in yield and/or reliability and/or efficiency, from the annealing process and/or the plasma treating process.

The data storage region **142***b* of the variable resistive material layer **142** may be a region in which a portion of the variable resistive material layer **142** is subjected to a plasma treatment process and/or an annealing process to decrease an oxygen vacancy concentration.

The data storage region **142***b* of the variable resistive material layer **142** may have different resistance, according to a set state and a reset state in operating the semiconductor device **100**. For example, among word lines WL, a program operation may turn off (OFF) selected word line WLa and may turn on (ON) unselected word lines WLb**1** and WL**1***b***2**. In this case, a current indicated by reference numeral CP in FIG. **3** may sequentially flow along a portion of the channel region **141***a* facing a first unselected word line WLb**1** located above the selected word line WLa, the selected word line WLa facing the selected word line WLa, and a portion of the channel region **141***a* facing a second unselected word line WLb**2** located below the selected word line WLa. A dotted line indicated by reference numeral CP in FIG. **3** may indicate a current flow during a program operation. For example, the current flow CP during the program operation may flow along the portion of the channel region **142***a* facing the first unselected word line WLb**1**, may shift to the data storage region **142***b* facing the selected word line WLa, may shift to the portion of the channel region **142***a* facing the second unselected word line WLb**2**, and may flow along the portion of the channel region **142***a*. As a current flows along the data storage region **142***b* facing the selected word line WLa, resistance of the data storage region **142***b* may be changed, and a portion of the data storage region **142***b* facing the selected word line WLa may be in a set state. By such a program operation, resistance of a portion of the data storage region **142***b* facing the selected word line WLa may be locally lowered.

An erase operation may turn off the selected word line WLa, similarly to the program operation, and may turn off the unselected word lines WLb**1** and WLb**2**, but may flow a current in a direction, opposite to the current flow during the above-described program operation, to change a magnetic field, to change a portion of the data storage region **142***b* facing the selected word line WLa in a reset state. Due to the erase operation, resistance of the portion of the data storage region **142***b* facing the selected word line WLa may be locally increased.

For example, the variable resistive material layer **142** may include at least one of hafnium oxide (HfO), zinc oxide (ZnO), indium oxide (InO), gallium oxide (GaO), tin oxide (SnO), copper oxide (CuO), molybdenum oxide (MoO), hafnium-silicon oxide (HSO), hafnium-zinc oxide (HZO), indium-zinc oxide (IZO), indium-gallium oxide (IGO), indium-tin oxide (ITO), indium-gallium-zinc oxide (IGZO), or indium-tin-zinc oxide (ITZO)

The core insulating pattern **145** may have a prismatic shape or a cylindrical shape extending in the vertical direction (Z). The core insulating pattern **145** may be disposed in a region including a center of the channel structure CH. An upper surface of the core insulating pattern **145** may be in contact with the pad pattern **149**. The core insulating pattern **145** may be formed of at least one of silicon oxide, silicon nitride, or silicon oxynitride.

The pad pattern **149** may be disposed on the core insulating pattern **145**, and may be in contact with an upper portion of the variable resistive material layer **142**. The pad pattern **149** may electrically connect the variable resistive material layer **142** to the bit lines **180**. The pad pattern **149** may be formed of doped polysilicon, for example, doped polysilicon having N-type conductivity.

The second separation pattern SS may extend between the first separation patterns MS in the X-direction. The second separation pattern SS may pass through the upper gate electrode **130**U, among the gate electrodes **130**, in the Z-direction, to separate them from each other in the Y-direction. The number of and/or thicknesses of upper gate electrodes **130**U, separated by the second separation pattern SS, may be variously changed in some example embodiments. The upper gate electrodes **130**U, separated by the second separation pattern SS, may form different string select lines. The second separation pattern SS may include an insulating material, for example, silicon oxide, silicon nitride, or silicon oxynitride.

The contact plugs **170** may be disposed between the channel structures CH and the bit lines **180**. The contact plugs **170** may be respectively connected to the channel pad **149**. The contact plugs **170** may be connected to the bit lines **180**. The contact plugs **170** may pass through at least one of the upper insulating layers **191** and **192**, for example, a first upper insulating layer **191** and a second upper insulating layer **192** in the Z-direction. In some example embodiments, a plurality of studs connected to the contact plugs **170** may be further disposed between one channel structure CH and one bit line **180**.

The contact plugs **170** may include a conductive pattern and a barrier layer covering side and bottom surfaces of the conductive pattern. The barrier layer may include, for example, at least one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), or tantalum nitride (TaN). The conductive pattern may include a metal material, for example, at least one of tungsten (W), titanium (Ti), copper (Cu), cobalt (Co), aluminum (Al), or alloys thereof. In some example embodiments, the contact plugs **170** may be formed as a plurality of plug structures.

The bit lines **180** may be disposed on the stacked structure ST and the channel structures CH, and may extend in the Y-direction. The bit lines **180** may be electrically connected to the circuit elements **20** of the first structure **1** through through-contact plugs. The bit lines **180** may be electrically connected to the variable resistive material layer **141**.

The bit lines **180** may include a conductive pattern and a barrier layer covering side and bottom surfaces of the conductive pattern. The barrier layer may include, for example, at least one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), or tantalum nitride (TaN). The conductive pattern may include a metal material, for example, at least one of tungsten (W), titanium (Ti), copper (Cu), cobalt (Co), aluminum (Al), or alloys thereof.

The upper insulating layers **191** and **192** may be disposed on the stacked structure ST. The upper insulating layers **191** and **192** may include a first upper insulating layer **191** and a second upper insulating layer **192**, sequentially stacked on the stacked structure ST. The upper insulating layers **191** and **192** may be formed of an insulating material such as silicon oxide.

FIGS. **4**A to **4**D are graphs illustrating an oxygen vacancy concentration in a variable resistive material layer of a semiconductor device according to various example embodiments.

Referring to FIG. **4**A, a variable resistive material layer **142** may have an oxygen vacancy concentration varying a concentration in a stepped profile, e.g. in a piecewise linear profile. For example, an oxygen vacancy concentration of the variable resistive material layer **142** in a width direction may decrease in a step profile, in a direction from a gate insulating layer **141** toward a core insulating pattern **145**. For example, a channel region **142***a* of the variable resistive material layer **142** may have an oxygen vacancy at a first concentration C**1**, and a data storage region **142***b* of the variable resistive material layer **142** may have an oxygen vacancy at a second concentration C**2**, lower than the first concentration C**1**. The first concentration C**1** may be a constant concentration according to a change in thickness of the channel region **142***a*, and the second concentration C**2** may be a constant concentration according to a change in thickness of the data storage region **142***b*.

Referring to FIGS. **4**B and **4**C, a variable resistive material layer **142** may have an oxygen vacancy concentration that gradually changes, e.g. that decreases in a polynomial or exponential manner or a piecewise polynomial or exponential manner, and an oxygen vacancy concentration in a data storage region **142***b* of the variable resistive material layer **142** may be lower than an oxygen vacancy concentration in a channel region **142***a* of the variable resistive material layer **142**.

Referring to FIG. **4**B, for example, the oxygen vacancy concentration in the channel region **142***a* of the variable resistive material layer **142** may increase as the channel region **142***a* approaches a gate insulating layer **141**, and the oxygen vacancy concentration in the data storage region **142***b* may decrease as the data storage region **142***b* approaches a core insulating pattern **145**. For example, in the channel region **142***a*, an oxygen vacancy concentration of a portion of the channel region **142***a* adjacent to the gate insulating layer **141** may be higher than an oxygen vacancy concentration of a portion of the channel region **142***a* adjacent to the data storage region **142***b*. For example, in the data storage region **142***b*, an oxygen vacancy concentration of a portion of the data storage region **142***b* adjacent to the channel region **142***a* may be higher than an oxygen vacancy concentration of a portion of the data storage region **142***b* adjacent to the core insulating pattern **145**.

Referring to FIG. **4**C, for example, an oxygen vacancy concentration of a channel region **142***a* of a variable resistive material layer **142** may decrease as the channel region **142***a* approaches a gate insulating layer **141**, and an oxygen vacancy concentration of the data storage region **142***b* may increase as the data storage region **142***b* approaches a core insulating pattern **145**. For example, in the channel region **142***a*, an oxygen vacancy concentration of a portion of the channel region **142***a* adjacent to the gate insulating layer **141** may be higher than an oxygen vacancy concentration of a portion of the channel region **142***a* adjacent to the data storage region **142***b*. For example, in the data storage region **142***b*, an oxygen vacancy concentration of a portion of the data storage region **142***b* adjacent to the channel region **142***a* may be lower than an oxygen vacancy concentration of a portion the data storage region **142***b* adjacent to the core insulating pattern **145**.

Referring to FIG. **4**D, a variable resistive material layer **142** may have an oxygen vacancy concentration that constantly changes. For example, in the variable resistive material layer **142**, the oxygen vacancy concentration may gradually decrease from a channel region **142***a* to a data storage region **142***b*. Therefore, the data storage region **142***b* may have a lower oxygen vacancy concentration, compared to the channel region **142***a*.

FIG. **5**A is a partially enlarged view of a portion of a semiconductor device according to various example embodiments.

FIG. **5**B is a graph illustrating an oxygen vacancy concentration in a variable resistive material layer of a semiconductor device according to various example embodiments.

Referring to FIGS. **5**A and **5**B, a data storage region **142***b* of a variable resistive material layer **142** may include a first data storage region **142***b***1** adjacent to a channel region **142***a* and a second data storage regions **142***b***2** adjacent to a core insulating pattern **145**. The first data storage region **142***b***1** may have oxygen vacancies at a second concentration C**2**, lower than a first concentration C**1** of the channel region **142***a*, and the second data storage region **142***b***2** may have oxygen vacancies at a third concentration C**3**, lower than the first concentration C**1** and higher than the second concentration C**2**. In FIG. **5**B, the variable resistive material layer **142** is illustrated as having an oxygen vacancy concentration varying a concentration in a stepped profile, but the variable resistance layer **142** may have an oxygen vacancy concentration that gradually changes, as illustrated in FIGS. **4**B and **4**C, or may have an oxygen vacancy concentration that constantly changes, as illustrated in FIG. **4**D.

FIGS. **6**A and **6**B are partially enlarged views illustrating a portion of a semiconductor device according to various example embodiments. FIGS. **6**A and **6**B illustrate a region corresponding to a region indicated by portion 'B' of FIG. **2**.

Referring to FIG. **6**A, a channel structure CHa may further include an epitaxial layer **107**. The epitaxial layer **107** may be disposed to contact an upper substrate **101** on a lower end of the channel structure Cha, and may be disposed adjacent to a side surface of at least one gate electrode **130**. The epitaxial layer **107** may be disposed in a recessed region of the upper substrate **101**. An upper surface of the epitaxial layer **107** may be higher than an upper surface of a lower gate electrode **130**L, and may be lower than the lower surface of a gate electrode **130** on the lower gate electrode **130**L, but the present inventive concept is not limited thereto. The epitaxial layer **107** may be connected to a variable resistive material layer **142** through the upper surface of the epitaxial layer **107**. An insulating layer **109** may be further disposed between the epitaxial layer **107** and the lower gate electrode **130**L adjacent to the epitaxial layer **107**.

Referring to FIG. **6**B, a semiconductor device **100** may further include a first horizontal conductive layer **102** disposed along an upper surface of an upper substrate **101** and a second horizontal conductive layer **103** extending along an upper surface of the first horizontal conductive layer **102**. The first horizontal conductive layer **102** and the second horizontal conductive layer **103** may be disposed between the upper substrate **101** and a stacked structure ST. At least a portion of the first horizontal conductive layer **102** and at least a portion of the second horizontal conductive layer **103** may be formed of polysilicon having N-type conductivity. The first horizontal conductive layer **102** may pass through a gate insulating layer **141** below a channel structure CHb, and may be in contact with a side surface of a variable resistive material layer **142**.

FIGS. **7** to **9** are schematic cross-sectional views of a semiconductor device according to various example embodiments.

Referring to FIG. **7**, gate electrodes **130** of a semiconductor device **100**A may include a first gate portion **131** adjacent to channel structures CH and a second gate portion **132** adjacent to first isolation patterns MS, respectively. The first gate portion **131** may surround side surfaces of the channel structures CH.

The first gate portion **131** may be formed of doped polysilicon, and the second gate portion **132** may be formed of a metal-semiconductor compound (e.g., one or more of WSi, TiSi, or the like), a metal nitride (e.g., WN, TiN, or the like), and/or a metal (e.g., W or the like).

Each of the gate electrodes **130** may include the first and second gate portions **131** and **132**, to improve electrical characteristics of the gate electrodes **130**. Therefore, in some example embodiments, a semiconductor device having improved electrical characteristics may be provided.

Referring to FIG. **8**, in a semiconductor device **100**B, a stacked structure ST of a second structure **2** may include a lower stacked structure and an upper stacked structure on the lower stacked structure, and channel structures CHc may include a lower channel structure passing through the lower stacked structure and an upper channel structure passing through the upper stacked structure, respectively. A variable resistive material layer **142** of the first channel structure and a variable resistive material layer **142** of the second channel structure may be connected to each other. In the connection region, a gate insulating layer **141** and a variable resistive material layer **142** may be respectively bent. For example, a side surface of the variable resistive material layer **142** may include a bent portion due to a difference in width in the connection region, and a side slope may be changed. Example embodiments a case in which a stacked structure is a double stacked structure, and inventive concepts may also include a multi-stacked structure that may be a double or more stacked structure. Furthermore a number of gate electrodes **130** below the bent portion may be the same as, or greater than, or less than a number of gate electrodes above the bent portion.

Referring to FIG. **9**, a first structure **1** and a second structure **2** of a semiconductor device **100**C may be bonded to each other through a bonding structure without a further adhesive layer. The second structure **2** of the semiconductor device **100**C is illustrated as vertically inverting the second structure **2** of the semiconductor device **100** of FIG. **2**. The semiconductor device **100**C may further include an upper bonding pad **165** and a lower bonding pad **65**. The second structure **2** may further include a third upper insulating layer **193**. The upper bonding pad **165** may be electrically connected to a bit line **180** through an upper bonding via **163**, and the lower bonding pad **65** may be electrically connected to circuit elements **20** through a lower via **63**. The lower bonding pad **65** and the upper bonding pad **165** may include, for example, tungsten (W), aluminum (Al), copper (Cu), tungsten nitride (WN), tantalum nitride (TaN), or titanium nitride (TiN), or a combination thereof, respectively. The lower bonding pad **65** and the upper bonding pad **165** may function as bonding layers for bonding the first structure **1** and the second structure **2**. In addition, the lower bonding pad **65** and the upper bonding pad **165** may provide an electrical connection path between the first structure **1** and the second structure **2**. The lower bonding pad **65** and the upper bonding pad **165** may be bonded by copper(Cu)-to-copper(Cu) bonding. Alternatively or in addition to the copper-to-copper bonding, the first structure **1** and the second structure **2** may be bonded by dielectric-to-dielectric bonding. The dielectric-to-dielectric bonding may form, for example, a portion of each of the third upper insulating layer **193** and a lower insulating layer **40**, and may be a bonding by dielectric layers surrounding the upper bonding pad **165** and the lower bonding pad **65**.

FIGS. **10**A to **11**B are flowcharts illustrating a method of manufacturing a semiconductor device according to various example embodiments.

FIGS. **12** to **23** are schematic views illustrating a method of manufacturing a semiconductor device according to various example embodiments.

Referring to FIGS. **10**A, **10**B, and **12**, a first structure **1** including circuit elements **20** and a lower interconnection structure **30** may be formed on a lower substrate **10**, an upper substrate **101** may be formed on the first structure **1**, sacrificial layers **110** and interlayer insulating layers **120** may be alternately stacked on the upper substrate **101** (S**10**), and openings OP passing through the sacrificial layers **110** and the interlayer insulating layers **120** may be formed (S**21**).

First, device isolation layers **15***s* may be formed in a lower substrate **10**, and a circuit gate dielectric layer **24** and a circuit gate electrode **26** may be sequentially formed on an active region **15***a*. The device isolation layers **15***s* may be formed by, for example, a shallow trench isolation (STI) process. The circuit gate dielectric layer **24** may be formed of silicon oxide, and the circuit gate electrode **26** may be formed as at least one of a polysilicon layer or a metal-semiconductor compound layer, but is not limited thereto. Next, a spacer layer **28** may be formed on both sidewalls of the circuit gate dielectric layer **24** and both sidewalls of the circuit gate electrode **26**, and source/drain regions **22** may be formed in the active region **15***a*. In some example embodiments, the spacer layer **28** may be formed as a plurality of layers. The source/drain regions **22** may be formed by performing an ion implantation process and/or an in-situ dopant deposition process; however, example embodiments are not limited thereto.

Lower contacts **32** and lower interconnections **34** of the lower interconnection structure **30** may be formed by partially forming a lower insulating layer **40**, etching and removing a portion thereof, and filling a conductive material therein, or may be formed by depositing a conductive material, patterning the same, and filling a portion removed by the patterning with a portion of a lower insulating layer **40**.

The lower insulating layer **40** may be formed as a plurality of insulating layers. The lower insulating layer **40** may be partially formed in each operation of forming the lower interconnection structure **30**, and may be further partially formed on an uppermost lower interconnection **34**, to be finally prepared to cover the circuit elements **20** and the lower interconnection structure **30**.

The upper substrate **101** may be formed of, for example, polysilicon. Polysilicon constituting the upper substrate **101** may include impurities, which may be implanted and/or may be incorporated during deposition of the polysilicon.

The sacrificial layers **110** may be partially replaced by a gate electrodes **130** (refer to FIG. **2**) by a subsequent process. The sacrificial layers **110** may be formed of a material, different from that of the interlayer insulating layers **120**, and may be formed of a material that may be etched with etching selectivity for the interlayer insulating layers **120** under specific etching conditions. For example, the interlayer insulating layer **120** may be formed of at least one of silicon oxide or silicon nitride, and the sacrificial layers **110** may be formed of a material, different from that of the interlayer insulating layer **120**, selected from silicon, silicon oxide, silicon carbide, and silicon nitride.

The sacrificial layer **110** may be referred to as a 'gate layer,' and when the sacrificial layer **110** includes polysilicon, the gate layer including polysilicon may function as a gate electrode, and a process of replacing the sacrificial layer **110** with a subsequent gate electrode **130** may be omitted. In some example embodiments, thicknesses of the interlayer insulating layers **120** may not all be the same. Thicknesses of the interlayer insulating layers **120** and the sacrificial layers **110** and/or the number of layers constituting the interlayer insulating layers **120** and the sacrificial layers **110** may be variously changed from those illustrated. A preliminary stacked structure may be formed by stacking the sacrificial layers **110** and the interlayer insulating layers **120**. After the preliminary stacked structure is formed, second separation patterns MS2 passing through a portion of upper sacrificial layers **110**, among the sacrificial layers **110**, may be formed.

Openings OP may be formed by anisotropically etching the preliminary stacked structure. Due to a height of the preliminary stacked structure, side surfaces of the openings OP may be inclined with respect to an upper surface of the upper substrate **101**.

Referring to FIGS. **10**B, **11**A, **13**A, and **14**, a gate insulating layer **141** covering the side surfaces of the openings OP may be formed (S**22**), and a variable resistive material layer **142**P may be formed. (S**23**), and the forming a variable resistive material layer **142**P (S**23**) may include forming a first variable resistive material layer **142_1** (S**23**A) and forming a second variable resistive material layer **142_2** on the first variable resistive material layer **142_1** (S**23**B).

The gate insulating layer **141** may be formed to conformally cover side and lower surfaces of the openings OP. The gate insulating layer **141** may be partially formed even on a level higher than an uppermost interlayer insulating layer **120**. After partially opening a lower portion of the gate insulating layer **141**, the first variable resistive material layer **142_1** and the second variable resistive material layer **142_2** may be conformally formed. The first variable resistive material layer **142_1** may be in contact with the upper substrate **101**, and may be in contact with the gate insulating layer **141**. The first variable resistive material layer **142_1** and the second variable resistive material layer **142_2** may be formed of the same material, and may be formed by, for example, performing an atomic layer deposition (ALD) process. The first variable resistive material layer **142_1** and the second variable resistive material layer **142_2** may be formed, for example, at the same time and/or in the same process chamber; however, example embodiments are not limited thereto. Although the first variable resistive material layer **141_1** and the second variable resistive material layer **142_2** are illustrated as respective layers for convenience of description, but may be substantially formed as a single layer (**142**P).

Referring to FIGS. **10**B, **11**B, **13**B, and **14**, a gate insulating layer **141** covering the side surfaces of the openings OP may be formed (S**22**), and a variable resistive material layer **142**P may be formed. (S**23**), and the forming a variable resistive material layer **142**P (S**23**) may include forming a variable resistive material capping layer 142' filling the opening OP and contacting the gate insulating layer 141 (S23A'), and etching the variable resistive material capping layer 142' (S23 B'). The gate insulating layer 141 may be formed to conformally cover side and lower surfaces of the openings OP. The gate insulating layer 141 may be partially formed even on a level higher than an uppermost interlayer insulating layer 120. After partially opening a lower portion of the gate insulating layer 141, for example, a chemical vapor deposition (CVD) process or a sputtering process may be performed to form the variable resistive material capping layer 142'. The variable resistive material capping layer 142' may fill the opening OP, and may be in contact with the gate insulating layer 141 in the opening OP. A central region of the variable resistive material capping layer 142' may be etched, to form the variable resistive material layer 142P extending along a side surface of the gate insulating layer 141 and having a specific thickness from the side surface of the gate insulating layer 141. The specific thickness may be predetermined, or alternatively may be dynamically or variably determined.

As in FIG. 13A-14, a method of forming the variable resistive material layer 142P before performing the following post-processing process, as described above, may include the manufacturing processes of FIGS. 13A and 14, or may include the manufacturing processes FIGS. 13B and 14.

Referring to FIGS. 10B and 14, a plasma treatment process PP may be performed on the variable resistive material layer 142P to change an oxygen vacancy concentration in a portion of the variable resistive material layer 142P (S24), to form a variable resistive material layer 142 including a channel region 142*a* and a data storage region 142*b*. The plasma treatment process PP may be performed using a source gas including one or more of oxygen (02), hydrogen (H_2), silane (SiH_4), argon (Ar), or the like. For example, a plasma treatment process PP using argon (Ar) may be performed, which may reduce an oxygen vacancy concentration in a portion of a surface of the variable resistive material layer 142P. Through this, a structure in which an oxygen vacancy concentration in the channel region 142*a* is relatively lower than an oxygen vacancy concentration in the data storage region 142*b* may be formed.

For example, an annealing process may be performed, instead of or in addition to the plasma processing process PP, to change an oxygen vacancy concentration in a portion of the variable resistive material layer 142P. For example, an annealing process may be performed to reduce an oxygen vacancy concentration in a region from a surface of the variable resistive material layer 142P. The annealing process may be or may include a thermal annealing process and/or a laser annealing process.

For example, referring to FIG. 13A together, before forming the first variable resistive material layer 142_1 and forming the second variable resistive material layer 142_2, a plasma processing process PP and/or an annealing process on the first variable resistive material layer 142_1 may be performed to change the oxygen vacancy concentration. Thereafter, after forming the second variable resistive material layer 142_2, an additional plasma treatment process PP and/or an additional annealing process may be performed or may not be performed.

A change in oxygen vacancy concentration in a width direction of the variable resistive material layer 142, e.g., an oxygen vacancy concentration profile or an oxygen vacancy concentration distribution, may be determined, for example, by one or more of Energy Dispersive X-ray Spectroscopy (EDS), X-ray Photoelectron Spectrometry (XPS), Secondary Ion Mass Spectrometry (SIMS), Rutherford back-scattering (RBS), Raman spectroscopy, Mott-Schottky impedance spectroscopy, x-ray crystallography (XRD), or the like.

Referring to FIGS. 10B and 16 to 19, a core insulating pattern 145 may be formed to cover the variable resistive material layer 142 and fill at least a portion of the opening OP (S25). The forming a core insulating pattern 145 (S25) may include forming a first core insulating layer 145A, partially removing the first core insulating layer 145A and the variable resistive material layer 142 from upper portions thereof, respectively, filling the opening OP with a second core insulating layer 145B, and forming the core insulating pattern 145 by partially removing the first and second core insulating layers 145A and 145B from upper portions thereof.

First, the first core insulating layer 145A may be formed to conformally cover an inner side surface of the variable resistive material layer 142 in the opening OP. Next, the first core insulating layer 145A and the variable resistive material layer 142 may be partially removed from upper portions thereof, to lower heights of upper ends thereof, compared to an upper end of the gate insulating layer 141. Next, the second core insulating layer 145B may fill an entirely unfilled space of the opening OP. An interface between the first core insulating layer 145A and the second core insulating layer 145B may be seen, depending on process conditions, but may not be clearly distinguished. Next, a planarization process such as a chemical mechanical planarization process and/or an etch-back process may be performed to remove a portion of the second core insulating layer 145B covering an upper end of the gate insulating layer 141. A space for forming a subsequent pad pattern 149 may be formed in the opening OP by partially removing the first and second core insulating layers 145A and 145B from upper portions thereof.

Referring to FIGS. 10A, 10B, 20, and 21, a pad pattern 149 may be formed on the core insulating pattern 145 (S26). The forming a pad pattern 149 (S26) may include forming a capping material layer 149P on the core insulating pattern 145, and performing a planarization process such as a chemical mechanical planarization process and/or an etch-back process to remove a portion of the capping material layer 149P disposed on the upper end of the gate insulating layer 141. During the planarization process, a portion of the gate insulating layer 141 on an uppermost interlayer insulating layer 120 may also be removed. Therefore, channel structures CH passing through the preliminary stacked structure may be formed (S20).

Referring to FIG. 22, separation openings T passing through the sacrificial layers 110 and the interlayer insulating layers 120 may be formed, and the sacrificial layers 110 may be removed through the separation openings T to form horizontal openings LT.

First, after the channel structure CH is formed, a first upper insulating layer 191 may be formed on the channel structure CH. The separation openings T may be formed by forming a mask layer and anisotropically etching the first upper insulating layer 191, the sacrificial layers 110, and the interlayer insulating layers 120, using a photolithography process. The separation openings T may be formed in a trench shape extending in the X-direction, and may expose the upper substrate 101 from lower ends of the separation openings T.

Next, the sacrificial layers 110 may be selectively removed with respect to the interlayer insulating layers 120 and the first upper insulating layer **191** through the separation openings T. Therefore, a plurality of horizontal openings LT may be formed between the interlayer insulating layers **120**.

Referring to FIGS. **10**A and **23**, gate electrodes **130** may be formed in the horizontal openings LT, and first separation patterns MS may be formed in the separation openings T (S**30**).

First, the gate electrodes **130** may be formed by filling the horizontal openings LT, formed by removing the sacrificial layers **110** through the separation openings T, with a conductive material. Therefore, a stacked structure ST in which the interlayer insulating layers **120** and the gate electrodes **130** are alternately stacked may be formed. The forming the gate electrodes **130** may include sequentially forming a first layer and a second layer.

Next, the first separation patterns MS may be formed by filling the separation openings T with an insulating material. In some example embodiments, a separation insulating pattern including an insulating material and a conductive core pattern including a conductive material may be sequentially formed in the separation openings T. The conductive core pattern may be formed to be spaced apart from the gate electrodes **130** and to contact the upper substrate **101**.

Next, a second upper insulating layer **192** may be formed, and contact plugs **170** and bit lines **180** may be formed (S**40**), to manufacture the semiconductor device **100** of FIGS. **1** to **3**.

FIG. **24** is a view schematically illustrating a data storage system including semiconductor devices according to various example embodiments.

Referring to FIG. **24**, a data storage system **1000** may include a semiconductor device **1100** and a controller **1200** electrically connected to the semiconductor device **1100**. The data storage system **1000** may be a storage device including one or more semiconductor devices **1100**, or an electronic device including the storage device. For example, the data storage system **1000** may be a solid state drive device (SSD), a universal serial bus (USB), a computing system, a medical device, or a communication device, including one or more semiconductor devices **1100**.

The semiconductor device **1100** may be or may include a non-volatile memory device, for example, a NAND flash memory device described above with reference to FIGS. **1** to **9**. The semiconductor device **1100** may include a first structure **1100**F, and a second structure **1100**S on the first structure **1100**F. In some example embodiments, the first structure **1100**F may be disposed next to the second structure **1100**S. The first structure **1100**F may be a peripheral circuit structure including a decoder circuit **1110**, a page buffer **1120**, and a logic circuit **1130**. The second structure **1100**S may be a memory cell structure including bit lines BL, a common source line CSL, word lines WL, first and second gate upper lines UL**1** and UL**2**, first and second gate lower lines LL**1** and LL**2**, and memory cell strings CSTR between each of the bit lines BL and the common source line CSL.

In the second structure **1100**S, each of the memory cell strings CSTR may include lower transistors LT**1** and LT**2** adjacent to the common source line CSL, upper transistors UT**1** and UT**2** adjacent to each of the bit lines BL, and a plurality of memory cell transistors MCT disposed between each of the lower transistors LT**1** and LT**2** and each of the upper transistors UT**1** and UT**2**. The number of lower transistors LT**1** and LT**2** and/or the number of upper transistors UT**1** and UT**2** may be variously changed according to various example embodiments, and may be the same, or different, from each other.

In some example embodiments, each of the upper transistors UT**1** and UT**2** may include a string select transistor, and each of the lower transistors LT**1** and LT**2** may include a ground select transistor. The lower gate lines LL**1** and LL**2** may be gate electrodes of the lower transistors LT**1** and LT**2**, respectively. The word lines WL may be gate electrodes of the memory cell transistors MCT, and the upper gate lines UL**1** and UL**2** may be gate electrodes of the upper transistors UT**1** and UT**2**, respectively.

In some example embodiments, the lower transistors LT**1** and LT**2** may include a lower erase control transistor LT**1** and a ground select transistor LT**2**, connected in series. The upper transistors UT**1** and UT**2** may include a string select transistor UT**1** and an upper erase control transistor UT**2**, connected in series. At least one of the lower erase control transistor LT**1** or the upper erase control transistor UT**2** may be used for an erase operation of erasing data stored in the memory cell transistors MCT using a gate-induced-drain-leakage (GIDL) phenomenon.

The common source line CSL, the first and second gate lower lines LL**1** and LL**2**, the word lines WL, and the first and second gate upper lines UL**1** and UL**2** may be electrically connected to the decoder circuit **1110** through first connection interconnections **1115** extending from the first structure **1100**F into the second structure **1100**S. The bit lines BL may be electrically connected to the page buffer **1120** through second connection interconnections **1125** extending from the first structure **1100**F into the second structure **1100**S.

In the first structure **1100**F, the decoder circuit **1110** and the page buffer **1120** may perform a control operation on at least one selected memory cell transistor among the plurality of memory cell transistors MCT. The decoder circuit **1110** and the page buffer **1120** may be controlled by the logic circuit **1130**. The semiconductor device **1100** may communicate with the controller **1200** through an input/output pad **1101** electrically connected to the logic circuit **1130**. The input/output pad **1101** may be electrically connected to the logic circuit **1130** through input/output connection interconnections **1135** extending from the first structure **1100**F into the second structure **1100**S.

The controller **1200** may include a processor **1210**, a NAND controller **1220**, and a host interface **1230**. According to some example embodiments, the data storage system **1000** may include a plurality of semiconductor devices **1100**, and in this case, the controller **1200** may control the plurality of semiconductor devices **1100**.

The processor **1210** may control an overall operation of the data storage system **1000** including the controller **1200**. The processor **1210** may operate according to a predetermined or alternatively variably determined firmware, and may access to the semiconductor device **1100** by controlling the NAND controller **1220**. The NAND controller **1220** may include a NAND interface **1221** processing communications with the semiconductor device **1100**. A control command for controlling the semiconductor device **1100**, data to be written to the memory cell transistors MCT of the semiconductor device **1100**, data to be read from the memory cell transistors MCT of the semiconductor device **1100**, or the like may be transmitted through the NAND interface **1221**. The host interface **1230** may provide a communication function between the data storage system **1000** and an external host. When a control command is received from the external host through the host interface **1230**, the processor **1210** may control the semiconductor device **1100** in response to the control command.

FIG. **25** is a perspective view schematically illustrating a data storage system including a semiconductor device according to various example embodiments.

Referring to FIG. **25**, a data storage system **2000** according to various example embodiments of inventive concepts may include a main substrate **2001**, a controller **2002** mounted on the main substrate **2001**, a semiconductor package **2003**, which may be provided as one or more semiconductor packages, and a DRAM **2004**. The semiconductor package **2003** and the DRAM **2004** may be connected to the controller **2002** by interconnection patterns **2005** formed on the main substrate **2001**.

The main substrate **2001** may include a connector **2006** including a plurality of pins, which may be coupled to an external host. The number and an arrangement of the plurality of pins in the connector **2006** may vary according to a communication interface between the data storage system **2000** and the external host. In some example embodiments, the data storage system **2000** may be communicated with the external host according to any one interface of a universal serial bus (USB), peripheral component interconnection express (PCI-Express), serial advanced technology attachment (SATA), M-Phy for universal flash storage (UFS), or the like. In some example embodiments, the data storage system **2000** may be operated by power supplied from the external host through the connector **2006**. The data storage system **2000** may further include a power management integrated circuit (PMIC) distributing power, supplied from the external host, to the controller **2002** and the semiconductor package **2003**.

The controller **2002** may write data to the semiconductor package **2003** or read data from the semiconductor package **2003**, and may improve an operation speed of the data storage system **2000**.

The DRAM **2004** may be or may include a buffer memory reducing a difference in speed between the semiconductor package **2003**, which may be a data storage space, and the external host. The DRAM **2004** included in the data storage system **2000** may also operate as a type of cache memory, and may provide a space temporarily storing data in a control operation on the semiconductor package **2003**. When the DRAM **2004** is included in the data storage system **2000**, the controller **2002** may further include a DRAM controller controlling the DRAM **2004** in addition to a NAND controller controlling the semiconductor package **2003**.

The semiconductor package **2003** may include first and second semiconductor packages **2003***a* and **2003***b*, spaced apart from each other. Each of the first and second semiconductor packages **2003***a* and **2003***b* may be a semiconductor package including a plurality of semiconductor chips **2200**. Each of the first and second semiconductor packages **2003***a* and **2003***b* may include a package substrate **2100**, semiconductor chips **2200** on the package substrate **2100**, adhesive layers **2300** disposed on a lower surface of each of the semiconductor chips **2200**, a connection structure **2400** electrically connecting each of the semiconductor chips **2200** and the package substrate **2100**, and a molding layer **2500** covering the semiconductor chips **2200** and the connection structure **2400** on the package substrate **2100**.

The package substrate **2100** may be a printed circuit board including package upper pads **2130**. Each of the semiconductor chips **2200** may include an input/output pad **2210**. The input/output pad **2210** may correspond to the input/output pad **1101** of FIG. **24**. Each of the semiconductor chips **2200** may include stacked structures **3210** and memory channel structures **3220**. Each of the semiconductor chips **2200** may include a semiconductor device according to any one of embodiments described above with reference to FIGS. **1** to **9**.

In some example embodiments, the connection structure **2400** may be a bonding wire electrically connecting the input/output pad **2210** and the upper package pads **2130**. Therefore, in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a bonding wire process, and may be electrically connected to the package upper pads **2130** of the package substrate **2100**. According to embodiments, in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a connection structure including a through silicon via (TSV), instead of a connection structure **2400** by a bonding wire process.

In some example embodiments, the controller **2002** and the semiconductor chips **2200** may be included in one (1) package. In some example embodiments, the controller **2002** and the semiconductor chips **2200** may be mounted on a further interposer substrate, different from the main substrate **2001**, and the controller **2002** and the semiconductor chips **2200** may be connected to each other by an interconnection formed on the interposer substrate.

FIG. **26** is a cross-sectional view schematically illustrating a semiconductor package according to various example embodiments. FIG. **26** may illustrate various example embodiments of the semiconductor package **2003** of FIG. **25**, and may conceptually illustrate a region taken along line II-II' of the semiconductor package **2003** of FIG. **25**.

Referring to FIG. **26**, in the semiconductor package **2003**, the package substrate **2100** may be a printed circuit board. The package substrate **2100** may include a package substrate body portion **2120**, package upper pads **2130** disposed on an upper surface of the package substrate body portion **2120** (see FIG. **25**), lower pads **2125** disposed on a lower surface of the package substrate body portion **2120** or exposed from the lower surface, and internal interconnections **2135** electrically connecting the upper pads **2130** and the lower pads **2125** in the package substrate body portion **2120**. The upper pads **2130** may be electrically connected to the connection structures **2400**. The lower pads **2125** may be connected to the interconnection patterns **2005** of the main substrate **2001** of the data storage system **2000**, as illustrated in FIG. **25**, through conductive connection portions **2800**.

Each of or at least some of the semiconductor chips **2200** may include a semiconductor substrate **3010**, and a first structure **3100** and a second structure **3200**, sequentially stacked on the semiconductor substrate **3010**. The first structure **3100** may include a peripheral circuit region including peripheral interconnections **3110**. The second structure **3200** may include a common source line **3205**, a stacked structure **3210** on the common source line **3205**, channel structures **3220** and separation regions **3230**, passing through the stacked structure **3210**, bit lines **3240** electrically connected to the memory channel structures **3220**, and gate contact plugs **3235** electrically connected to word lines WL (refer to FIG. **24**) of the stacked structure **3210**. As described above with reference to FIGS. **1** to **9**, each of the semiconductor chips **2200** may include a lower substrate **10**, circuit elements **20**, an upper substrate **101**, gate electrodes **130**, channel structures CH, first separation patterns MS, second separation patterns SS, and bit lines **180**.

Each of or at least some of the semiconductor chips **2200** may include a through-interconnection **3245** electrically connected to the peripheral interconnections **3110** of the first structure **3100** and extending into the second structure **3200**. The through-interconnection **3245** may disposed outside the stacked structure **3210**, and may be further disposed to pass through the stacked structure **3210**. Each of the semiconductor chips **2200** may further include an input/output pad **2210** electrically connected to the peripheral interconnections **3110** of the first structure **3100** (refer to FIG. **25**).

A method of manufacturing a semiconductor device having improved electrical characteristics and a simplified manufacturing process, in which a variable resistive material layer includes a channel region having a high oxygen vacancy concentration and a data storage region having a low oxygen vacancy concentration, may be provided.

Various advantages and effects of inventive concepts are not limited to the above, and will be more easily understood in the process of describing specific example embodiments of inventive concepts.

Any of the elements and/or functional blocks disclosed above may include or be implemented in processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc. The processing circuitry may include electrical components such as at least one of transistors, resistors, capacitors, etc. The processing circuitry may include electrical components such as logic gates including at least one of AND gates, OR gates, NAND gates, NOT gates, etc.

While various example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of inventive concepts as defined by the appended claims. Furthermore example embodiments are not necessarily mutually exclusive with one another. For example, some example embodiments may include one or more features described with reference to one or more figures, and may also include one or more other features described with reference to one or more other figures.

What is claimed is:

1. A method of manufacturing a semiconductor device, comprising:

forming a stacked structure by stacking gate layers and interlayer insulating layers alternately on a substrate; and

forming a channel structure that passes through the stacked structure in a vertical direction,

wherein the forming a channel structure includes,

forming an opening by etching the stacked structure,

forming a gate insulating layer covering at least a side surface of the opening,

forming a variable resistive material layer on the gate insulating layer,

changing an oxygen vacancy concentration in a region of the variable resistive material layer by performing one or both of a plasma treatment process on the variable resistive material layer or an annealing process on the variable resistive material layer,

forming a core insulating pattern covering the variable resistive material layer and filling at least a portion of the opening, after performing the one or both of plasma treatment process or the annealing process, and

forming a pad pattern on the core insulating pattern.

2. The method of claim **1**, wherein the variable resistive material layer is formed by the one or both of the plasma treatment process or the annealing process to include a channel region including oxygen vacancies at a first concentration, and a data storage region including oxygen vacancies at a second concentration, less than the first concentration.

3. The method of claim **2**, wherein the channel region of the variable resistive material layer is formed to contact the gate insulating layer and to extend along a side surface of the gate insulating layer, and

the data storage region of the variable resistive material layer is formed to contact the core insulating pattern and to extend along a side surface of the core insulating pattern.

4. The method of claim **2**, wherein the data storage region of the variable resistive material layer is formed to be farther from the side surface of the opening, compared to the channel region.

5. The method of claim **1**, wherein the variable resistive material layer is formed of a transition metal oxide.

6. The method of claim **1**, wherein the variable resistive material layer is formed of at least one of hafnium oxide (HfO), zinc oxide (ZnO), indium oxide (InO), gallium oxide (GaO), tin oxide (SnO), copper oxide (CuO), molybdenum oxide (MoO), hafnium-silicon oxide (HSO), hafnium-zinc oxide (HZO), indium-zinc oxide (IZO), indium-gallium oxide (IGO), indium-tin oxide (ITO), indium-gallium-zinc oxide (IGZO), or indium-tin-zinc oxide (ITZO).

7. The method of claim **1**, wherein the annealing process includes one or both of a thermal annealing process or a laser annealing process.

8. The method of claim **1**, wherein the forming a variable resistive material layer on the gate insulating layer comprises:

forming a first layer including a transition metal oxide on the gate insulating layer; and

forming a second layer including a transition metal oxide on the first layer.

9. The method of claim **8**, wherein the one or both of plasma treatment process or the annealing process is performed on at least one of the first layer or the second layer.

10. The method of claim **9**, wherein an oxygen vacancy concentration in the transition metal oxide of the first layer increases in the first layer by the plasma treatment process.

11. The method of claim **9**, wherein an oxygen vacancy concentration in the transition metal oxide of the second layer decreases in the second layer by the annealing process.

12. The method of claim **1**, wherein the forming a variable resistive material layer on the gate insulating layer comprises:

forming a variable resistive material capping layer filling the opening; and

etching a central region of the variable resistive material capping layer to form the variable resistive material layer extending along a side surface of the gate insulating layer and having a specific thickness from the side surface of the gate insulating layer.

13. The method of claim **1**, wherein the forming a core insulating pattern comprises:

forming a first core insulating layer conformally on the variable resistive material layer;

removing a portion of the first core insulating layer and a portion of the variable resistive material layer from upper portions thereof, respectively;

forming a second core insulating layer filling the opening and contacting the first core insulating layer; and

forming the core insulating pattern by removing a portion of the first core insulating layer and a portion of the second core insulating layer from upper portions thereof, and

the forming a pad pattern comprises:

forming a pad space by removing an upper portion of the core insulating pattern; and

forming the pad pattern by depositing a conductive material in the pad space.

14. A method of manufacturing a semiconductor device, comprising:

forming a stacked structure by stacking gate layers and interlayer insulating layers alternately on a substrate; and

forming a channel structure passing through the stacked structure in a vertical direction,

wherein the forming a channel structure includes,

forming an opening by etching the stacked structure,

forming a gate insulating layer covering at least a side surface of the opening;

forming a variable resistive material layer on the gate insulating layer and including a first region and a second region,

changing an oxygen vacancy concentration in either of the first region or the second region,

forming a core insulating pattern filling at least a portion of the opening, and

forming a pad pattern on the core insulating pattern.

15. The method of claim **14**, wherein the variable resistive material layer is formed of a transition metal oxide.

16. The method of claim **14**, wherein the changing an oxygen vacancy concentration in either of the first region or the second region comprises decreasing an oxygen vacancy concentration in the second region, compared to an oxygen vacancy concentration in the first region,

wherein the first region is in contact with the gate insulating layer, and

the second region is in contact with the core insulating pattern.

17. The method of claim **14**, wherein the variable resistive material layer has an oxygen vacancy concentration profile gradually decreasing along a width of the variable resistive material layer in a direction from the gate insulating layer toward a central region of the channel structure.

18. The method of claim **14**, further comprising:

forming a separation opening passing through the stacked structure;

removing the gate layers through the separation opening; and

forming gate electrodes by filling regions from which the gate layers are removed with a conductive material.

19. A method of manufacturing a semiconductor device, comprising:

forming a stacked structure by stacking gate layers and interlayer insulating layers alternately on a substrate; and

forming a channel structure passing through the stacked structure in a vertical direction,

wherein the forming a channel structure comprises,

forming an opening by etching the stacked structure,

forming a gate insulating layer in the opening,

forming a variable resistive material capping layer filling the opening and contacting the gate insulating layer,

etching a central region of the variable resistive material capping layer to form a variable resistive material layer extending along a side surface of the gate insulation layer and having a specific thickness from the side surface of the gate insulation layer,

changing an oxygen vacancy concentration in a region of the variable resistive material layer by performing one or both of a plasma treatment process on the variable resistive material layer or an annealing process on the variable resistive material layer;

forming a core insulating pattern that covers the variable resistive material layer and that fills at least a portion of the opening, after performing the one or both of the plasma treatment process or the annealing process; and

forming a pad pattern on the core insulating pattern.

20. The method of claim **19**, wherein the annealing process includes one or both of a thermal annealing process or a laser annealing process,

the variable resistive material layer is formed by the plasma treatment process or the annealing process to include a channel region including oxygen vacancies at a first concentration, and a data storage region including oxygen vacancies at a second concentration, less than the first concentration,

wherein the channel region is formed in a portion of the variable resistive material layer contacting the gate insulating layer, and

the data storage region is formed in a portion of the variable resistive material layer contacting the core insulating pattern.

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